
FLOW-ALIGNED DISTILLATION: PRESERVING LAYER-WISE RESIDUAL TRAJECTORIES IN COMPRESSED LANGUAGE MODELS

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ABSTRACT

A large portion of the parameters in a transformer are allocated by feed-forward network (FFN) layers. Compressing a transformer via FFN accordingly alters the residual update at each layer, and the resulting local mismatches can propagate through depth. The evolution of the residual stream across the transformer layers is viewed as a *trajectory* in layers. Within each block, the post-attention state is linearly interpolated with the next-layer hidden state. The derivative along this interpolation is constant and equals the FFN residual update, which is seen as the local velocity of the transition. Flow-aligned distillation (FAD) is proposed by aligning these local velocities between the teacher and the student. FAD projects these velocities into a low-rank coordinate system estimated from frozen teacher velocities, where the projected deviations are using the inverse square root of a layer-wise diagonal covariance estimated from the teacher. Starting from a variational lower bound on mutual information, this paper presents a maximum-entropy conditional model whose conditional mean is the projected student velocity and whose covariance is formed by the projected teacher velocities. Solving this problem corresponds to maximizing a Gaussian conditional or minimizing a Mahalanobis alignment loss. The coordinate system is therefore estimated from the teacher and is derived through the relative inverse-variance weighting across projected directions. To investigate the effect of the distillation objective, we compare competing losses in identical architectures with shared FFN weights, training data, and optimization settings. Using Llama-3.2-3B and Llama-3.1-8B, FAD outperforms distillation at the logit-level and hidden-state matching. It also improves over matched-compression baselines while reducing model parameters by 20.93% at the primary operating point. These results support aligning FFN-induced local transitions along the layer-wise trajectory as an effective principle for capability recovery in structurally compressed language models.

1 INTRODUCTION

Structural compression reduces the model storage, but also perturbs the computation performed at each layer. In a residual transformer, such perturbations can propagate through depth because each block consumes the state produced by the preceding blocks. In previous methods (Romero et al., 2015; Jiao et al., 2020; Wang et al., 2020; Sanh et al., 2019; Gu et al., 2024; Agarwal et al., 2024), the output-logit distillation might constrain the final prediction, whereas the hidden-state matching might constrain the representations reached in the selected layers. Neither of them directly supervised the feed-forward network (FFN) update that transformed the post-attention state into the next-layer hidden state. We therefore argue whether the local quantity in a compressed student should be reproduced to follow the teacher’s computation across layers.

Consequently, we view the evolution of the residual stream across transformer layers as a trajectory in the layer-wise direction. Within each block, the residual FFN update connects the post-attention state to the next-layer hidden state. The derivative of the corresponding straight-line interpolation is constant and equals this update, giving it a local velocity interpretation. This motivates aligning teacher and student FFN velocities rather than matching only the final outputs or the hidden states.

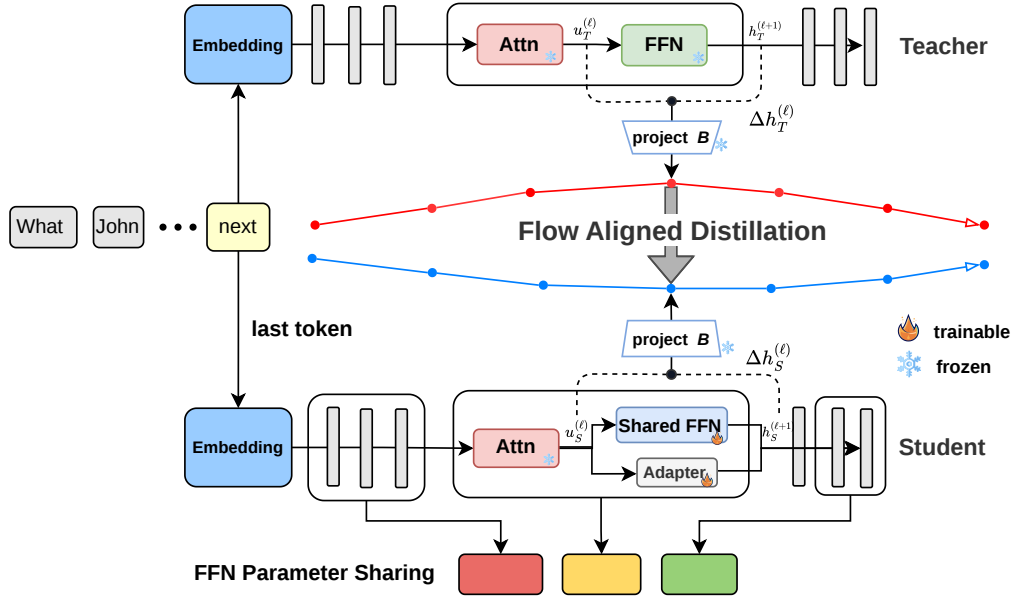


Figure 1: Overview of flow-aligned distillation (FAD) with three stages. The frozen teacher (top) and the compressed student (bottom) process the same input, and at the last prediction token T , each block ℓ exposes its FFN residual update as a one-step transport velocity, $v_T^{(\ell)} = \Delta h_T^{(\ell)}$ and $v_S^{(\ell)} = \Delta h_S^{(\ell)}$. (a) Attention is frozen in both models; teacher keeps its L independent FFNs, whereas student replaces them with *shared* weights together with a trainable per-layer low-rank adapter by Eq.(13), and layers with similar FFN inputs are tied to a single core (the colored “Shared FFN Parameter” blocks: red, yellow and green indicate three parts of sharing). (b) A shared low-rank teacher basis (“project B ”) maps both velocities to their subspace codes $\{z_T^{(\ell)}, z_S^{(\ell)}\}$ by Eq.(3). (c) Flow-aligned distillation pulls the student transport (blue) onto the teacher transport (red) layer by layer; the training signal is the teacher-induced information-theoretic objective, which is arranged as FAD loss by Eq.(11).

This paper presents flow-aligned distillation (FAD), which compares these velocities in a geometry estimated from the frozen teacher. FAD projects teacher and student velocities into teacher-derived low-rank coordinates and accounts for their layer-wise directional scales. A variational mutual-information bound (Barber & Agakov, 2003; Poole et al., 2019), together with a maximum-entropy conditional model (Jaynes, 1957), yields a Gaussian conditional which results in a Mahalanobis alignment objective. To evaluate the distillation signal, all compared methods use the same shared-weight student architecture, frozen teacher, training data, and optimization protocol. There are three-fold findings which are presented to

- interpret the transformer’s FFN residual updates as the one-step local transport velocities and formulate the shared-FFN compression as the preservation of teacher transport information under a parameter budget,
- combine the teacher-derived low-rank coordinates with the layer-wise directional normalization to obtain the FAD objective,
- match or even outperform the performance of teacher model using the compressed FAD models where FFN parameters are compressed, and validates over different information-theoretic transport targets, teacher-derived transport coordinates, and teacher-induced scale matrix.

2 RELATED WORK

Structural compression and parameter sharing. Quantization (Jacob et al., 2018; Frantar et al., 2023; Lin et al., 2024), pruning (Han et al., 2016; Frantar & Alistarh, 2023; Sun et al., 2024; Ma et al., 2023), layer removal (Men et al., 2025; Chen et al., 2025; Tian et al., 2026; An et al., 2024) and weight tying (Lan et al., 2020; Dehghani et al., 2019; Zhussip et al., 2026; Smith et al., 2025) reduce the model cost through different structural constraints. Their optimization criteria are developed

primarily through parameter or activation reconstruction or saliency, or end-to-end task loss. These criteria can preserve the overall quality of the model, but none of them explicitly supervises the residual update produced by each compressed layer. The proposed FAD aims to provide a layer-wise recovery signal for a fixed compressed architecture, which consolidates FFN operators based on shared structural banks, leaving attention and logical depth unchanged.

Intermediate knowledge distillation. Knowledge distillation is commonly performed for model compression by matching output distributions, hidden states, attention relations, or student-generated sequences (Romero et al., 2015; Jiao et al., 2020; Wang et al., 2020; Sanh et al., 2019; Gu et al., 2024; Agarwal et al., 2024). This study similarly preserves the intermediate variables. Rather than matching a complete hidden state, this paper matches the FFN residual update that mimics the teacher’s residual trajectory. It also separates coordinate selection from error scaling: both models are projected into teacher-derived velocity coordinates, and their deviations are standardized by the layer-wise teacher covariance. Where a feature-matching loss using its weights is usually chosen and tuned, here all three choices including variable, coordinate, and scale follow from the same variational objective.

Flow matching and information bottleneck. Flow matching is feasible to enable learning by aligning the velocity along a prescribed path (Chen et al., 2018; Lipman et al., 2023; Liu et al., 2023). This study would like to capture the geometric information from a teacher based on the realized residual increment at each layer in a way that a student is learned for information preservation. In (Barber & Agakov, 2003; Poole et al., 2019), the variational mutual-information bounds were derived to validate the information bottleneck, which is referred to illustrate the proposed FAD for the compression-prediction trade-off between teacher velocities as the source information, student velocities as the target representation, and a finite rank budget forcing the dominant teacher directions to be sufficiently preserved.

3 FLOW-ALIGNED DISTILLATION

This paper deals with model compression by aligning the teacher and student large language models (LLMs) based on their FFN residual updates in a low-rank geometry built by the frozen teacher. We first define the aligned variable, then construct its teacher-derived coordinates and directional scales, and finally derive the alignment objective. The shared-weight student is implemented in the fixed structural-compression setting. However, the derived FAD objective does not depend on a particular parameter-sharing assignment. Throughout, L denotes the number of layers and D the dimension of the features. The subscripts T and S indicate the teacher and the student, respectively. The map $c : [L] \rightarrow [Q]$ assigns a layer to a coordinate of Q groups, while $g : [L] \rightarrow [K]$ assigns it to the K parameter-sharing groups. Velocities are extracted at the final non-padding input position used to predict the next token.

3.1 FFN RESIDUAL UPDATE AS LOCAL VELOCITY

For the distillation model $M \in \{T, S\}$ and the input sequence x , let $\mathcal{H}_M(x) = (h_M^{(1)}, \dots, h_M^{(L+1)})$ denote its discrete layer-wise trajectory of hidden states $h_M^{(\ell)}$. The ℓ -th pre-norm block or layer first produces the post-attention state $u_M^{(\ell)}$ and then updates it as $\Delta h_M^{(\ell)}$ through the FFN

$$u_M^{(\ell)} = h_M^{(\ell)} + \text{Attn}_M^{(\ell)}(\text{Norm}_1(h_M^{(\ell)})), \quad \Delta h_M^{(\ell)} = \text{FFN}_M^{(\ell)}(\text{Norm}_2(u_M^{(\ell)})), \quad h_M^{(\ell+1)} = u_M^{(\ell)} + \Delta h_M^{(\ell)} \quad (1)$$

where two normalizations are performed on $h_M^{(\ell)}$ and $u_M^{(\ell)}$, respectively. The residual update of FFN $\Delta h_M^{(\ell)}$ is therefore the displacement from the post-attention state $u_M^{(\ell)}$ to the hidden state of the next-layer $h_M^{(\ell+1)}$. In particular, the layer-wise flow transition $\phi_M^{(\ell)}(t)$, where $t \in [0, 1]$, is represented by a linear interpolation, where its local velocity $v_M^{(\ell)}$ is defined by differentiation as follows

$$\phi_M^{(\ell)}(t) = u_M^{(\ell)} + t\Delta h_M^{(\ell)}, \quad v_M^{(\ell)} \triangleq \frac{\partial \phi_M^{(\ell)}(t)}{\partial t} = \Delta h_M^{(\ell)}. \quad (2)$$

FAD aligns the velocities $v_T^{(\ell)}$ and $v_S^{(\ell)}$ at the supervised layers. Their mismatch changes the next-layer hidden state and can therefore propagate to subsequent blocks.

3.2 TEACHER-INDUCED COORDINATE AND SCALE

Having induced $\mathbf{v}_M^{(\ell)}$ as the local quantity to be aligned, we next evaluate how teacher-student velocity deviations are measured. An unweighted Euclidean loss in $\mathbb{R}^D$ treats individual full-space directions equally, despite the strongly non-uniform variations of teacher FFN updates. This method runs two operations: selecting a low-rank coordinate system and standardizing deviations within that system.

Teacher-induced coordinate. Layers with similar teacher-velocity statistics are assigned to one of the Q coordinate groups through $c : [L] \rightarrow [Q]$. For each group c , a coordinate based on the rank- r orthonormal PCA basis is constructed through a projection $\mathbf{B}_c \in \mathbb{R}^{D \times r}$ with teacher mean $\boldsymbol{\mu}_c \in \mathbb{R}^D$ from frozen teacher velocities $\mathbf{v}_{T,n}^{(\ell)}$ of the sample n . Teacher and student velocities are projected using the same basis and centroid

$$\mathbf{z}_{M,n}^{(\ell)} = \mathbf{B}_{c(\ell)}^\top (\mathbf{v}_{M,n}^{(\ell)} - \boldsymbol{\mu}_{c(\ell)}) \in \mathbb{R}^r, \quad M \in \{T, S\}. \quad (3)$$

This projection restricts the alignment objective to the leading directions of the teacher-velocity variations. The same centroid is applied to both models, resulting in the projected deviation

$$\mathbf{z}_{S,n}^{(\ell)} - \mathbf{z}_{T,n}^{(\ell)} = \mathbf{B}_{c(\ell)}^\top (\mathbf{v}_{S,n}^{(\ell)} - \mathbf{v}_{T,n}^{(\ell)}).$$

Layer-wise normalization. The projected teacher velocities still have substantially different scales or variances in coordinates and layers. Without normalization, high-variance coordinates would result in the loss and its gradients. We therefore estimate the layer-wise diagonal covariance matrix

$$(\boldsymbol{\sigma}_T^{(\ell)})^2 = \text{Var}_n[\mathbf{z}_{T,n}^{(\ell)}], \quad \Sigma_T^{(\ell)} = \text{Diag}((\boldsymbol{\sigma}_T^{(\ell)})^2) + \epsilon I_r \succ 0 \quad (4)$$

where I_r is the identity matrix, $\epsilon > 0$ prevents division by zero and ensures that $\Sigma_T^{(\ell)}$ is positive definite. In alignment loss, projected deviations are normalized by $(\Sigma_T^{(\ell)})^{-1/2}$, which inducing the precision metric $(\Sigma_T^{(\ell)})^{-1}$. Thus, the PCA basis determines which teacher-velocity directions are modeled, while the layer-wise covariance determines how deviations along those directions are weighted. All teacher-derived statistics are estimated once and kept fixed during student training.

3.3 FROM INFORMATION PRESERVATION TO FLOW ALIGNMENT

For each supervised layer $\ell \in \mathcal{S}$ from the teacher, let $p_\ell(\mathbf{z}_T^{(\ell)}, \mathbf{z}_S^{(\ell)})$ denote the joint distribution of the projected teacher and student velocities induced by the training data. FAD estimates the shared-weight student θ_S^* under Θ_{shared} that maximizes mutual information (MI) $I(\cdot)$ between projected teacher and student velocities and minimizes Kullback-Leibler (KL) divergence using layer-wise distribution $p_\ell(\cdot)$

$$\theta_S^* = \arg \max_{\theta_S \in \Theta_{\text{shared}}} \frac{1}{|\mathcal{S}|} \sum_{\ell \in \mathcal{S}} \left[I_{p_\ell}(\mathbf{z}_T^{(\ell)}; \mathbf{z}_S^{(\ell)}) - \mathbb{E}_{p_\ell(\mathbf{z}_S^{(\ell)})} [\text{KL}(p_\ell(\mathbf{z}_T^{(\ell)} | \mathbf{z}_S^{(\ell)}) \| q_{\theta_S^*}^*(\mathbf{z}_T^{(\ell)} | \mathbf{z}_S^{(\ell)})] \right] \quad (5)$$

where the conditional variational model $q_{\theta_S^*}^*$ is calculated with the highest entropy $H_q(\cdot)$ via

$$q_{\theta_S^*}^* = \arg \max_{q \in \mathcal{Q}_{\theta_S^*}^{(\ell)}} H_q(\mathbf{z}_T^{(\ell)} | \mathbf{z}_S^{(\ell)}), \quad \forall \ell \in \mathcal{S}. \quad (6)$$

In Eqs. (5) and (6), the student velocities are favored to preserve information about the paired teacher velocities. The KL term measures the gap between the real posterior $p_\ell(\cdot)$ and the variational posterior $q_{\theta_S^*}(\cdot)$ associated with the current student. The conditional model is specified without introducing an auxiliary inference network. The admissible variational densities satisfy

$$\mathcal{Q}_{\theta_S^*}^{(\ell)} = \left\{ q : \begin{aligned} & q(\mathbf{z}_T^{(\ell)} | \mathbf{z}_S^{(\ell)}) \geq 0, \quad \int q(\mathbf{z}_T^{(\ell)} | \mathbf{z}_S^{(\ell)}) d\mathbf{z}_T^{(\ell)} = 1, \\ & \mathbb{E}_q[\mathbf{z}_T^{(\ell)} | \mathbf{z}_S^{(\ell)}] = \mathbf{z}_S^{(\ell)}, \quad \text{Cov}_q[\mathbf{z}_T^{(\ell)} | \mathbf{z}_S^{(\ell)}] = \Sigma_T^{(\ell)} \end{aligned} \right\}. \quad (7)$$

The conditional mean is centered at the projected student velocity $\mathbf{z}_S^{(\ell)}$, while the covariance matrix is fixed to the directional scale of the layer-wise $\Sigma_T^{(\ell)}$ estimated by the teacher, which is an imposed modeling constraint rather than an estimate of the unknown conditional covariance matrix.

For any admissible conditional density, the Barber-Agakov identity (Barber & Agakov, 2003; Poole et al., 2019) is used to re-arrange the combined MI and KL terms in Eq. (5) to yield

$$H_{p_\ell}(\mathbf{z}_T^{(\ell)}) + \mathbb{E}_{p_\ell}[\log q_{\theta_S}^*(\mathbf{z}_T^{(\ell)}|\mathbf{z}_S^{(\ell)})]. \quad (8)$$

Because the teacher model θ_T and its projection $\mathbf{B}$ are frozen, $H_{p_\ell}(\mathbf{z}_T^{(\ell)})$ is independent of the student parameters θ_S . Maximizing Eq. (5) therefore reduces to maximizing the conditional log-likelihood of the paired teacher velocities $\mathbf{z}_T^{(\ell)}$. The expectation is evaluated directly from teacher-student pairs, so the real conditional density is not required to be estimated explicitly. Under the constraints in Eq. (7), the maximum-entropy principle for conditional density (Jaynes, 1957) is obtained by

$$q_{\theta_S}^*(\mathbf{z}_T^{(\ell)}|\mathbf{z}_S^{(\ell)}) = \mathcal{N}(\mathbf{z}_T^{(\ell)}; \mathbf{z}_S^{(\ell)}, \Sigma_T^{(\ell)}). \quad (9)$$

Its negative log-likelihood is then calculated to find the loss

$$-\log q_{\theta_S}^*(\mathbf{z}_T^{(\ell)}|\mathbf{z}_S^{(\ell)}) = \frac{1}{2} \left\| (\Sigma_T^{(\ell)})^{-1/2} (\mathbf{z}_S^{(\ell)} - \mathbf{z}_T^{(\ell)}) \right\|_2^2 + C_T^{(\ell)} \quad (10)$$

where $C_T^{(\ell)}$ is a log determinant depending only on the frozen teacher variance. Thus, each projected deviation is measured relative to the teacher variance of the corresponding direction. Removing the student-independent constant and absorbing the factor $\frac{1}{2}$ into the global loss coefficient gives

$$\mathcal{L}_{\text{FAD}} = \frac{1}{|S|} \sum_{\ell \in S} \mathbb{E}_n \left[\left\| (\Sigma_T^{(\ell)})^{-1/2} (\mathbf{z}_{S,n}^{(\ell)} - \mathbf{z}_{T,n}^{(\ell)}) \right\|_2^2 \right] \quad (11)$$

which is calculated as an expectation over all samples n . The student θ_S is trained by

$$\underset{\theta_S \in \Theta_{\text{shared}}}{\text{minimize}} \quad \mathcal{L}_{\text{train}} = \mathcal{L}_{\text{CE}} + \lambda_{\text{FAD}} \mathcal{L}_{\text{FAD}}. \quad (12)$$

The teacher covariance matrix determines the relative weighting across projected directions, while λ_{FAD} controls the overall strength of FAD relative to cross-entropy training through $\mathcal{L}_{\text{CE}}$.

3.4 SHARED-WEIGHT STUDENT FOR CONTROLLED EVALUATION

To isolate the distillation objective from the compression policy, all objective comparisons use the same shared-weight student. The teacher’s L independent FFNs are replaced by $K \ll L$ sets of shared FFN weights under assignment $g : [L] \rightarrow [K]$, while each layer maintains a private parallel low-rank adapter. Let $\xi_S^{(\ell)} = \text{Norm}_2(u_S^{(\ell)})$ denote the input to the student FFN. Its residual update is

$$\Delta h_S^{(\ell)} = \underbrace{F(\xi_S^{(\ell)}; \Theta_{\text{FFN}}^{(g(\ell))})}_{\text{shared-weight FFN}} + \underbrace{U^{(\ell)} V^{(\ell)} \xi_S^{(\ell)}}_{\text{private layer adapter}} \quad (13)$$

with the adapter parameters $V^{(\ell)} \in \mathbb{R}^{\rho \times D}$, $U^{(\ell)} \in \mathbb{R}^{D \times \rho}$, and $\rho = 128$. The shared FFN weights $\Theta_{\text{FFN}}^{(g(\ell))}$ capture the computation reused across layers, while the private adapter provides a layer-specific correction. We zero-initialize $U^{(\ell)}$, so the adapter initially contributes zero and each layer begins with only its assigned shared FFN weights. Attention, normalization, residual connections, and logical depth remain unchanged. Since both branches contribute directly to $\Delta h_S^{(\ell)}$, FAD supervises the complete compressed FFN update. The assignment g is computed offline by using an end-to-end functional replacement test, followed by the complete-link agglomerative grouping into exactly K groups and medoid initialization. Across all objective comparisons, g , K , ρ , and the initialization process are fixed. Only the distillation objective changes.

4 EXPERIMENTS

We organize the experiments around four questions. First, under an identical compressed architecture, does recovering the training tasks also preserve broader pretrained capability, and is any gain consistent across knowledge domains? Second, do teacher-velocity projection, coordinate selection, and layer-conditioned scaling make separate contributions? Third, does the resulting method remain competitive against complete structural-compression systems across model scales? Finally, does FAD measurably preserve the directional and scale-sensitive transport geometry that motivates it, and where does this task-aligned advantage cease to hold?

4.1 EXPERIMENTAL PROTOCOL

Data, teachers, and evaluation. We train on `commonsense_170k` (161,899 training and 8,521 validation examples). Each backbone is paired with a LoRA-fine-tuned and merged teacher (Hu et al., 2022). The in-domain score is the macro average over PIQA, Social-IQA, WinoGrande, ARC-Challenge, ARC-Easy, HellaSwag, and OpenBookQA (Bisk et al., 2020; Sap et al., 2019; Sakaguchi et al., 2021; Clark et al., 2018; Zellers et al., 2019; Mihaylov et al., 2018). To test broader capability rather than recovery-set fit, we evaluate all 14,042 MMLU questions (Hendrycks et al., 2021) with `lm_eval v0.4.12` under common zero- and five-shot protocols.

Controlled objective study. The primary analysis uses Llama-3.2-3B with $K = 19$ stored FFN banks, corresponding to a 20.93% reduction in non-embedding model parameters. Every row uses the same teacher, sharing assignment g , shared-weight student, private adapters, initialization, training data, checkpoint rule, and optimization protocol; only the recovery objective changes. We compare cross-entropy alone, logit KL, hidden-state MSE, ambient FFN-velocity MSE, isotropic matching in the rank-256 teacher-velocity subspace, and FAD. Every teacher-guided variant retains the same cross-entropy term and replaces only the auxiliary recovery term; CE-only sets the auxiliary coefficient to zero. Thus Table 1 is a matched intervention on the signal distilled into one fixed compressed model.

External compression study. We additionally compare with FLAP (An et al., 2024), Tyr-the-Pruner (Li et al., 2025), and LLM-Streamline (Chen et al., 2025) on Llama-3.2-3B and Llama-3.1-8B. These results form a separate matched-compression study and are not mixed with the controlled $K = 19$ objective ablation. The primary controlled objective study reports three independently trained seeds (42, 43, and 44). Other analyses remain single-run unless explicitly stated otherwise. Appendix A gives accounting and protocol details, and Appendix B reports the full task breakdown.

4.2 TASK RECOVERY IS NOT CAPABILITY PRESERVATION

Table 1: **Controlled recovery-objective comparison at the primary $K = 19$ operating point.** Architecture, parameter reduction, teacher, data, initialization, and optimization are fixed; only the recovery objective changes. Values are mean $\pm$ sample standard deviation over three independently trained seeds (42, 43, and 44). Best mean values in each column are bold.

Recovery objective	Commonsense $\uparrow$	MMLU 0-shot $\uparrow$	MMLU 5-shot $\uparrow$
CE only	85.10 $\pm$ 0.57	33.37 $\pm$ 5.34	34.08 $\pm$ 4.47
Logit KL	84.91 $\pm$ 0.08	39.88 $\pm$ 1.12	43.38 $\pm$ 0.47
Hidden-state MSE	85.06 $\pm$ 0.07	44.26 $\pm$ 2.21	45.71 $\pm$ 2.31
Ambient velocity MSE	85.17 $\pm$ 0.39	40.99 $\pm$ 1.47	43.50 $\pm$ 1.42
Projected isotropic velocity	84.96 $\pm$ 0.06	45.02 $\pm$ 0.72	48.60 $\pm$ 1.12
FAD	84.84 $\pm$ 0.42	47.55 $\pm$ 3.84	52.09 $\pm$ 1.96

Table 1 shows that the seven-task in-domain commonsense macro is largely insensitive to the recovery objective: the six objectives span only 0.33 points in mean accuracy, from 84.84 to 85.17. MMLU, however, more clearly separates the recovery objectives. FAD reaches 47.55 $\pm$ 3.84 zero-shot and 52.09 $\pm$ 1.96 five-shot accuracy, compared with 45.02 $\pm$ 0.72 and 48.60 $\pm$ 1.12 for the strongest non-FAD objective, projected isotropic velocity matching. This corresponds to average improvements of 2.53 and 3.49 points, respectively. Moreover, FAD achieves higher raw MMLU accuracy than projected isotropic matching in all three training seeds. These results indicate that the main advantage of FAD is not improved fit to the in-domain commonsense tasks, but stronger retention of MMLU capability under the same compressed architecture.

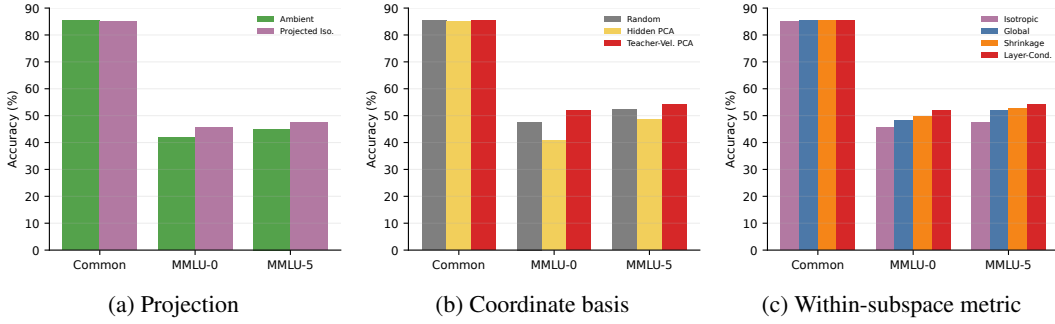


Figure 2: **Matched component interventions.** (a) Ambient versus projected isotropic velocity matching. (b) Random, hidden-state-PCA, and teacher-velocity-PCA coordinates. (c) Isotropic, global, shrinkage, and layer-conditioned metrics. All settings use the same compressed architecture and report commonsense and zero-/five-shot MMLU.

Table 2: **Cross-domain consistency of the primary MMLU gain.** Each entry reports zero-shot/five-shot accuracy. The strongest non-FAD value is selected separately in each domain and protocol from the controlled objectives in Table 1.

Domain	Strongest non-FAD	FAD	FAD gain
Humanities	41.87 / 43.04	46.31 / 49.76	+4.44/ + 6.72
Other	52.85 / 53.88	58.64 / 59.64	+5.79/ + 5.76
Social Sci.	53.56 / 56.19	61.88 / 63.44	+8.32/ + 7.25
STEM	40.66 / 42.50	44.15 / 46.02	+3.49/ + 3.52
Overall	45.73 / 47.64	51.97 / 54.10	+6.24/ + 6.46

Table 2 shows that the advantage is not produced by one favorable MMLU category. FAD improves every domain under both prompting protocols, with its smallest domain margin still 3.49/3.52 points and its largest in Social Sciences at 8.32/7.25 points. The full objective-by-domain matrix is reported in Appendix B.2.

4.3 DISSECTING THE FAD GAIN

Figure 2 separates three questions that would be confounded in the final FAD model. First, replacing ambient matching by projected isotropic matching changes commonsense by only -0.50 points but improves MMLU by $+3.96/+2.54$ points, showing that restricting the recovery signal to a compact subspace is useful. Second, within a teacher-aware scaling protocol, the teacher-velocity basis reaches $51.97/54.10$ MMLU, compared with $47.62/52.37$ for a random subspace and $40.85/48.49$ for hidden-state PCA. Random projection is competitive, so the supported claim is not that arbitrary projection fails; rather, projection supplies useful capacity allocation and teacher-velocity coordinates give the strongest overall retention. Third, within the fixed velocity subspace, the layer-conditioned metric improves MMLU by $+6.24/+6.52$ points over isotropic matching, while global and shrinkage estimators lie between the two. Projection determines *where* limited student capacity is spent, whereas the teacher scale determines *how* deviations inside that space are priced. Complete coordinate and metric tables are provided in Appendix C.

The rank-256 teacher basis also remains concentrated beyond calibration: it captures 45.20% of held-out PIQA velocity energy versus 8.25% for a random subspace, and retains about a $2.5\times$ advantage on WikiText-2 and C4. Because PCA maximizes teacher variance by construction, this is a structural diagnostic; Figure 2 provides the causal downstream evidence, and Appendix C.2 reports the full values.

Table 3: **External compression comparisons.** Panel (a) reports the seven-task commonsense macro at the nominal matched operating points used by the separate 3B/8B study. Panel (b) reports MMLU at the primary 3B operating point; reductions are the exact values associated with each run. Full per-task results and accounting definitions are in Appendix B.

(a) Commonsense macro across backbones					
Model	Compression rate	FLAP	Tyr	Streamline	FAD
Llama-3.1-8B	15%	79.83	83.61	86.86	87.67
Llama-3.1-8B	20%	76.99	81.72	86.87	87.66
Llama-3.2-3B	15%	70.59	79.30	81.29	85.38
Llama-3.2-3B	25%	63.89	72.71	81.25	85.03

(b) MMLU at the primary 3B point				
Method	Compression rate	0-shot	5-shot	
Dense teacher	0.00%	49.42	51.20	
FLAP	20.05%	35.29	36.36	
Tyr-the-Pruner	20.00%	40.31	40.41	
LLM-Streamline	18.80%	47.30	49.22	
FAD	20.93%	51.97	54.10	

4.4 COMPARISON WITH STRUCTURAL COMPRESSION BASELINES

FAD has the highest commonsense macro at all four 3B/8B operating points in Table 3(a). The margin over LLM-Streamline is modest on 8B (0.81/0.79 points) and larger on 3B (4.09/3.78 points), indicating that the recovery signal is especially useful under the tighter small-model bottleneck. At the primary 3B point, FAD also exceeds LLM-Streamline by 4.67/4.88 MMLU points while using the larger reported reduction. These comparisons complement, rather than replace, the controlled objective study: the external table tests the complete compression systems, whereas Table 1 isolates the recovery signal. The nominal 3B “25%” row belongs to the external-baseline experiment family and is not the controlled $K = 17$ checkpoint used in the compression stress test; the two rows must not be interpreted as repeated runs at one exact reduction.

4.5 TRANSPORT FIDELITY UNDER RECOVERY OBJECTIVES

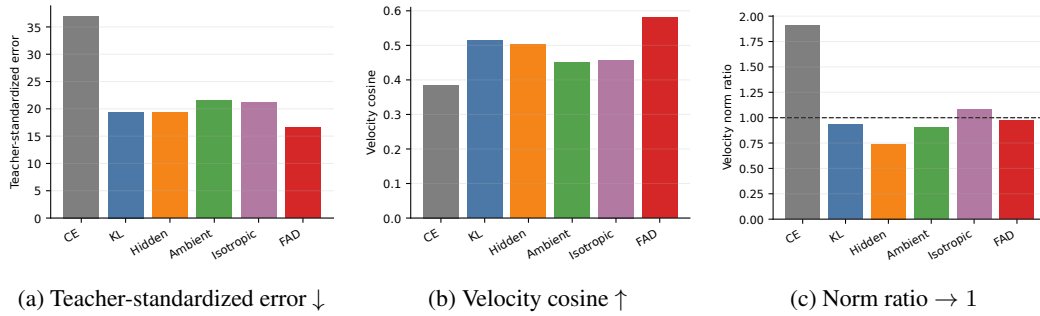


Figure 3: **Held-out transport diagnostics on PIQA.** FAD has the lowest teacher-standardized transport error, the highest velocity cosine, and the norm ratio closest to one. Hidden-state MSE attains smaller raw state and ambient velocity errors; the full diagnostic table is reported in Appendix C.2.

Figure 3 connects the capability result to the quantity FAD is designed to preserve. Its advantage is not a generic reduction of every Euclidean discrepancy: hidden-state matching has smaller raw state and ambient velocity errors. Instead, FAD most strongly preserves direction and magnitude in the frozen teacher geometry, which is the selective transport-preservation claim made by the method. The directional cosine is not rescaled by the FAD precision matrix and therefore provides a complementary check rather than merely restating the optimized quadratic loss.

4.6 SCOPE UNDER DISTRIBUTION SHIFT AND STRONGER COMPRESSION

The task-aligned geometry does not preserve generic autoregressive likelihood to the same degree: C4 NLL is 14.01 for FAD, 2.76 for the teacher, and 9.02 for logit KL. Low-frequency C4 replay lowers FAD NLL to 6.40 while retaining 85.40 commonsense and 52.82 five-shot MMLU, but does not restore dense-model likelihood. Thus FAD is a *task-aligned capability-preservation objective*, not a distribution-universal guarantee; Appendix E gives the full analysis.

The advantage also narrows when structural capacity dominates. At 25.58% and 30.23% reduction, FAD remains strong for commonsense and relative to ambient matching but is not the best MMLU objective. Appendix D reports this stress test: recovery supervision cannot fully compensate for an excessively restricted function class.

4.7 LIMITATIONS

The central $K = 19$ objective comparison uses three independently trained seeds per objective. Evaluation-sample uncertainty is reported separately from training-seed variability. We test two Llama scales under one shared-FFN testbed. The frozen geometry is calibration-dependent and transfers incompletely to generic text. Static sharing reduces stored parameters but not logical depth; latency remains implementation-dependent.

5 CONCLUSION

We studied which signal should be preserved after structural compression. Across three independently trained seeds, competing recovery objectives achieve similar in-domain commonsense accuracy, whereas FAD provides the strongest average MMLU retention by aligning local FFN transport in teacher-derived coordinates. Matched interventions isolate separate gains from projection, velocity coordinates, and layer-conditioned scaling, while held-out diagnostics confirm stronger directional and teacher-weighted transport fidelity. Gains extend to 3B/8B structural baselines, while generic-text and high-compression tests delimit the claim to task-aligned capability preservation.

REPRODUCIBILITY STATEMENT

Section 4.1 defines the protocols. Appendices A–E provide provenance, accounting, full results, and failure analyses; Appendix F fixes sharing; and Appendices G–H give the complete derivations and theoretical scope.

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Appendix organization. Appendix A specifies implementation and accounting; Appendix B gives full capability results; Appendix C reports component and trajectory analyses; Appendix D studies stronger compression; Appendix E diagnoses generic-language degradation; Appendix F documents the fixed sharing construction; Appendix G gives the complete variational derivation; and Appendix H states the geometric, optimization, and trajectory-stability results.

A IMPLEMENTATION AND REPRODUCIBILITY

A.1 RUN PROVENANCE

The primary controlled study is locked to the Llama-3.2-3B $K = 19$ bundle and its frozen rank-256 velocity geometry. The reported anchor is 51.97/54.10 MMLU under zero-/five-shot evaluation; these values are two protocols applied to the same provenance-locked model, not independent retraining runs. Internal run tags are not used as model names in the paper. The anonymous artifact should expose a single immutable manifest containing the exact model and tokenizer revisions, teacher checkpoint, geometry checksum, sharing assignment, adapter rank, optimizer configuration, and evaluation commit.

Table A.1: Primary controlled-study identifiers and settings supported by the reported results. Low-level optimizer and checkpoint metadata are stored in the submitted run manifest rather than reconstructed from prose.

Quantity	Primary value
Backbone	Llama-3.2-3B ($L = 28$)
Stored FFN banks	$K = 19$
Parameter reduction	20.93% of non-embedding model parameters
Projection variable	FFN residual velocity at the final valid context position
Projection rank	$r = 256$
Private adapter rank	$\rho = 128$
Training data	commonsense_170k: 161,899/8,521 train/validation
MMLU evaluator	lm_eval v0.4.12, 14,042 examples, 0-/5-shot
Evaluation seed	44
Primary MMLU anchor	51.97/54.10

A.2 TEACHERS, OPTIMIZATION, AND ARTIFACT CONTRACT

Each teacher is obtained by LoRA fine-tuning and merging. To avoid silently inventing run metadata, the exact LoRA targets, rank, scaling, dropout, base-model revision, optimizer, learning-rate schedule, warmup, weight decay, gradient clipping, effective batch size, sequence length, training steps, checkpoint rule, FAD coefficient, coordinate grouping, stabilizer, and calibration size must be read from the immutable anonymous configuration shipped with the corresponding row. The artifact contract is one configuration and one result manifest per reported model, together with frozen teacher geometry and scripts that regenerate all tables and figures. No table in this paper combines checkpoints whose bundle, geometry, or evaluator provenance differs from its stated experiment family.

A.3 EVALUATION PROTOCOL

Commonsense evaluation uses log-probability multiple-choice scoring with one common prompt and normalization protocol for all methods. MMLU uses one task revision, all 14,042 questions, seed 44, and identical zero-/five-shot settings. Geometry analyses use held-out examples and do not reuse training losses as downstream metrics. Any evaluation-example bootstrap interval must be reported separately from variability across independently trained models.

A.4 COMPRESSION ACCOUNTING

We distinguish stored banks K , FFN-only reduction including adapters, exact run-specific reduction, and rounded levels used only in the external 3B/8B study. The primary $K = 19$ point removes 20.93% of non-embedding parameters, and cross-family comparisons use exact reductions.

B FULL CAPABILITY RESULTS

B.1 FULL STRUCTURAL-BASELINE BREAKDOWN

Table B.1: **Seven-task matched-compression comparison.** Values are percent accuracy. These nominal operating points belong to the separate external compression study; they are not the controlled $K = 19$ objective ablation.

Method	Level	PIQA	SIQA	Wino	ARC-C	ARC-E	HellaS.	OBQA	Avg.
Llama-3.1-8B									
Dense teacher	–	88.57	79.79	83.66	80.38	90.91	95.45	84.40	86.17
FLAP	15%	83.08	76.66	80.43	68.60	81.86	92.40	75.80	79.83
Tyr-the-Pruner	15%	87.81	76.15	81.45	75.00	87.42	94.25	83.20	83.61
LLM-Streamline	15%	89.28	79.53	86.03	80.12	91.79	95.87	85.40	86.86
FAD	15%	89.72	80.66	86.35	81.40	92.68	96.08	86.80	87.67
FLAP	20%	81.39	75.74	75.37	64.76	77.36	90.88	73.40	76.99
Tyr-the-Pruner	20%	86.89	75.95	79.79	71.67	84.89	93.22	79.60	81.72
LLM-Streamline	20%	89.39	79.63	85.87	79.95	91.29	95.75	86.20	86.87
FAD	20%	89.77	80.71	86.74	81.66	92.72	96.00	86.00	87.66
Llama-3.2-3B									
Dense teacher	–	85.15	79.12	81.37	72.44	84.60	92.84	80.60	82.30
FLAP	15%	78.51	72.06	66.46	55.29	69.70	85.34	66.80	70.59
Tyr-the-Pruner	15%	83.41	76.31	77.58	67.41	80.72	90.49	79.20	79.30
LLM-Streamline	15%	84.22	77.02	80.27	72.18	84.01	91.32	80.00	81.29
FAD	15%	87.05	81.06	86.42	76.28	87.50	94.55	84.80	85.38
FLAP	25%	72.96	69.04	62.75	47.70	57.70	77.90	59.20	63.89
Tyr-the-Pruner	25%	78.62	72.42	71.27	57.25	71.84	84.98	72.60	72.71
LLM-Streamline	25%	83.90	76.61	80.35	71.93	84.64	91.91	79.40	81.25
FAD	25%	85.96	79.02	86.74	77.30	88.68	94.12	83.40	85.03

B.2 MMLU DOMAINS UNDER CONTROLLED OBJECTIVES

Table B.2: MMLU domain accuracy for the controlled $K = 19$ objective study. Each cell is zero-shot / five-shot. FAD is best in every domain and protocol.

Objective	Humanities	Other	Social Sci.	STEM	Overall
CE only	27.84/30.52	26.39/30.00	27.07/33.18	28.96/32.89	27.60/31.52
Logit KL	34.81/36.81	46.99/50.31	47.64/49.66	36.63/38.50	40.73/42.99
Hidden MSE	39.64/41.68	52.85/53.88	53.01/55.74	40.66/42.50	45.72/47.64
Ambient	36.90/40.45	47.34/51.14	47.29/51.84	38.19/39.26	41.77/45.04
Isotropic	41.87/43.04	50.53/52.66	53.56/56.19	39.14/40.95	45.73/47.58
FAD	46.31/49.76	58.64/59.64	61.88/63.44	44.15/46.02	51.97/54.10

Table 2 summarizes the strongest alternative in each domain. This full matrix shows which objective supplies that alternative and confirms that the overall gain is not driven by one domain.

B.3 MMLU DOMAINS AGAINST STRUCTURAL BASELINES

(a) 0-shot					(b) 5-shot				
Method	STEM	Hum.	Social	Other	Method	STEM	Hum.	Social	Other
FAD	44.15	46.31	61.88	58.64	FAD	46.02	49.76	63.44	59.64
Streamline	38.66	43.95	53.92	54.59	Streamline	41.33	44.44	56.68	57.06
Tyr	34.92	37.41	44.07	46.44	Tyr	35.65	36.81	46.15	45.00
FLAP	32.13	32.65	38.64	39.20	FLAP	32.98	33.50	40.10	40.39

Table B.3: MMLU domain-level accuracy against structural compression baselines.

B.4 TRAINING-SEED EVIDENCE

Table B.4: Multi-seed results for FAD at the primary operating point. Values report three independently trained seeds and their mean $\pm$ sample standard deviation.

Seed	Commonsense macro	MMLU 0-shot	MMLU 5-shot
42	84.47	44.99	50.19
43	84.76	45.70	51.99
44	85.30	51.97	54.10
Mean $\pm$ sample std.	84.84 $\pm$ 0.42	47.55 $\pm$ 3.84	52.09 $\pm$ 1.96

C COMPONENT, GEOMETRY, AND TRAJECTORY ANALYSES

C.1 COORDINATE AND METRIC CONTROLS

(a) Coordinate basis with teacher-aware scaling				(b) Metric inside velocity PCA			
Basis	Common.	M0	M5	Metric	Common.	M0	M5
Random subspace	85.28	47.62	52.37	Isotropic	85.03	45.73	47.58
Hidden-state PCA	85.01	40.85	48.49	Global teacher	85.38	48.22	51.83
Velocity PCA	85.30	51.97	54.10	Shrinkage teacher	85.31	49.60	52.66
				Layer-conditioned	85.30	51.97	54.10

Table C.1: Coordinate and metric controls at the primary operating point. Random projection is competitive, so the supported claim is that projection itself is useful and teacher-velocity coordinates with layer conditioning give the best overall retention—not that every random projection fails.

Table C.2: Held-out energy captured by the rank-256 teacher-velocity PCA basis and a random subspace of the same rank. This table establishes persistent low-rank teacher structure; downstream causality is tested by the matched interventions in Figure 2.

Held-out data	Teacher PCA	Random	Ratio
PIQA	45.20%	8.25%	5.48 $\times$
WikiText-2	20.83%	8.30%	2.51 $\times$
C4	20.42%	8.30%	2.46 $\times$

C.2 COMPLETE TRANSPORT DIAGNOSTICS

Table C.3: Aggregate held-out PIQA diagnostics corresponding to Figure 3. Final d_h is measured at the last block; the remaining quantities are averaged over the evaluated transport layers.

Recovery objective	Final $d_h \downarrow$	Mean $d_v \downarrow$	Teacher error $\downarrow$	Cosine $\uparrow$	Norm ratio $\rightarrow 1$
Hidden-state MSE	77.08	5.79	19.33	0.503	0.739
Ambient velocity MSE	90.65	6.63	21.53	0.450	0.901
FAD	84.64	6.97	16.60	0.581	0.972

The diagnostics separate raw Euclidean matching from teacher-weighted transport preservation. Hidden-state MSE minimizes final-state and ambient-velocity error, whereas FAD produces the lowest teacher-standardized error, strongest directional agreement, and most accurate update magnitude. This distinction prevents the mechanistic analysis from claiming universal dominance over every discrepancy.

C.3 PROJECTION RANK AND CALIBRATION SENSITIVITY

(a) Projection rank				(b) Geometry calibration	
r	Common.	M0	M5	Setting	Commonsense
32	85.13	37.38	43.07	1k examples	85.21
64	85.24	43.95	50.31	4k examples	85.32
128	85.57	46.40	49.22	16k examples	85.47
256	85.30	51.97	54.10	Full, seed 42	85.40
				Full, seed 43	85.22
				Full, seed 44	85.30

Table C.4: Projection-rank and calibration sensitivity. MMLU is not monotone in rank, so the supported statement is that $r = 256$ is best among the tested ranks, not that larger rank is guaranteed to improve performance.

C.4 TEACHER SCALE

The diagonal teacher scale ranges from approximately 8.0×10^{-5} to 74.8 (about 5.97 orders of magnitude), with layer-wise condition numbers from roughly 82 to 3,845. This anisotropy motivates layer-wise standardization; the matched metric intervention in the main text establishes its downstream contribution.

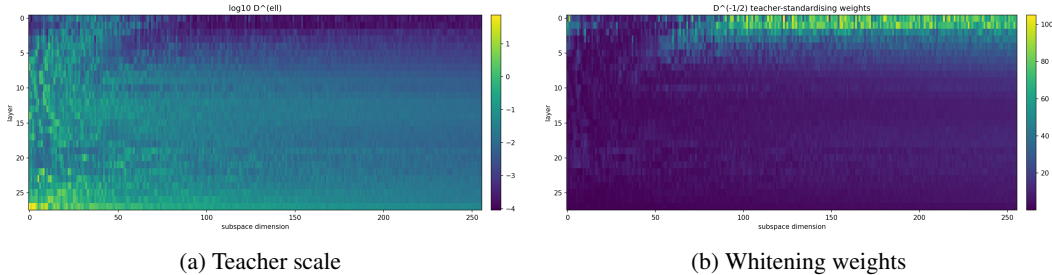


Figure C.1: Layer-by-direction teacher scale and corresponding standardization weights with identical ordering. The broad range motivates direction-dependent normalization; Figure 2(c) tests its downstream contribution.

D COMPRESSION-REGIME ROBUSTNESS

D.1 OBJECTIVE COMPARISON UNDER STRONGER COMPRESSION

Table D.1: Controlled recovery objectives across compression regimes. Best values within each K block are bold.

K	Reduction	Objective	Common.	MMLU-0	MMLU-5
19	20.93%	Logit KL	84.97	40.73	42.99
19	20.93%	Hidden MSE	84.98	45.72	47.64
19	20.93%	Ambient	85.53	41.77	45.04
19	20.93%	Isotropic	85.03	45.73	47.58
19	20.93%	FAD	85.30	51.97	54.10
17	25.58%	Logit KL	77.37	36.36	39.02
17	25.58%	Hidden MSE	77.74	38.63	36.38
17	25.58%	Ambient	76.31	30.12	27.15
17	25.58%	Isotropic	77.36	30.27	31.23
17	25.58%	FAD	78.69	37.46	31.95
15	30.23%	Logit KL	71.11	28.85	31.49
15	30.23%	Hidden MSE	70.01	28.27	29.25
15	30.23%	Ambient	67.87	23.24	23.04
15	30.23%	Isotropic	71.81	28.78	29.08
15	30.23%	FAD	73.32	26.88	28.44

FAD has the strongest commonsense recovery at $K = 17$ and $K = 15$ and remains substantially better than ambient matching. It is not the strongest MMLU objective in these aggressively compressed regimes, and the absolute $K = 15$ MMLU score is near the lower end of the evaluation range. These results support robustness to ambient matching, not a claim that FAD dominates every objective at every capacity.

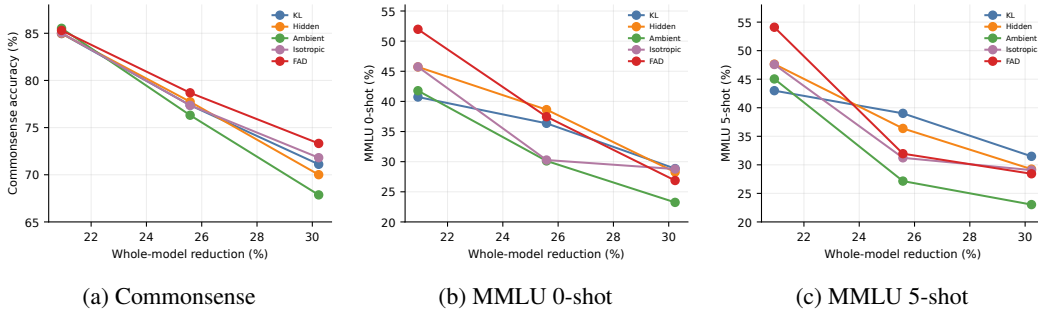


Figure D.1: Controlled objectives under stronger structural compression. The primary claim is anchored at $K = 19$; $K = 17/15$ expose the capacity-limited regime.

D.2 NESTED STRUCTURAL RESTRICTION

For fixed teacher geometry and a refinement sequence of sharing partitions, Theorem 9 orders the global optima of the nested model classes. The observed FAD loss rises from 0.104 to 0.176 as exact reduction increases from 13.95% to 30.23% for $K = 22, 19, 17, 15$.

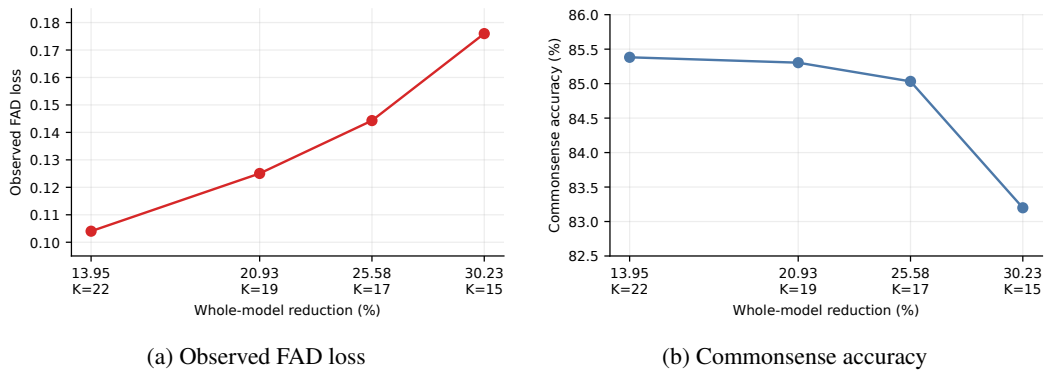


Figure D.2: Nested structural restriction and observed transport distortion. The monotone trained losses support the nested-class prediction but do not prove convergence of the nonconvex CE–FAD training problem.

E GENERIC-LANGUAGE TRADE-OFF AND MITIGATION

E.1 EVALUATOR SANITY CHECK

Table E.1: Uncompressed generic-language likelihood verifies that the evaluator is operating in a normal range before structural recovery.

Model	Wiki NLL	Wiki PPL	C4 NLL	C4 PPL
Base model	2.233	9.33	2.456	11.66
Teacher	2.518	12.41	2.760	15.80

E.2 GENERIC-LANGUAGE DEGRADATION BY OBJECTIVE

Table E.2: C4 likelihood under the controlled recovery objectives. Lower is better. All structurally recovered variants degrade, while all-token logit KL is the least affected.

Objective	C4 NLL	C4 PPL
CE only	15.963	8,566,947
Logit KL	9.024	8,298
Hidden MSE	12.030	167,631
Ambient	13.535	755,494
Isotropic	14.091	1,316,761
FAD	14.011	1,215,258

The recovery data emphasize commonsense decisions and the aligned signal is read at the final prediction position, so generic all-token language modeling is underconstrained. The result is not unique to FAD, but it rules out a universal language-model preservation claim.

E.3 LOW-FREQUENCY GENERIC REPLAY

Table E.3: Mitigating the generic-language trade-off. Mild C4 replay gives the best observed balance, but remains far from the uncompressed likelihood.

Variant	Common.	M0	M5	Wiki NLL	C4 NLL	C4 PPL
FAD anchor	85.30	51.97	54.10	13.121	14.011	1,215,258
FAD + KL(.25)	85.07	41.34	52.07	10.149	10.574	39,111
FAD + mild replay (1/10)	85.40	47.22	52.82	6.687	6.398	600.8
FAD + strong replay (1/4)	85.27	45.64	52.67	7.534	7.189	1,324.6
FAD + KL + replay	84.87	41.61	52.08	7.375	7.474	1,761.9

Mild replay lowers C4 NLL by 7.61 (54.3%), leaves commonsense unchanged, and retains 52.82 five-shot MMLU. Its C4 perplexity remains much higher than the base model and teacher, and the remedy evidence is currently single-seed.

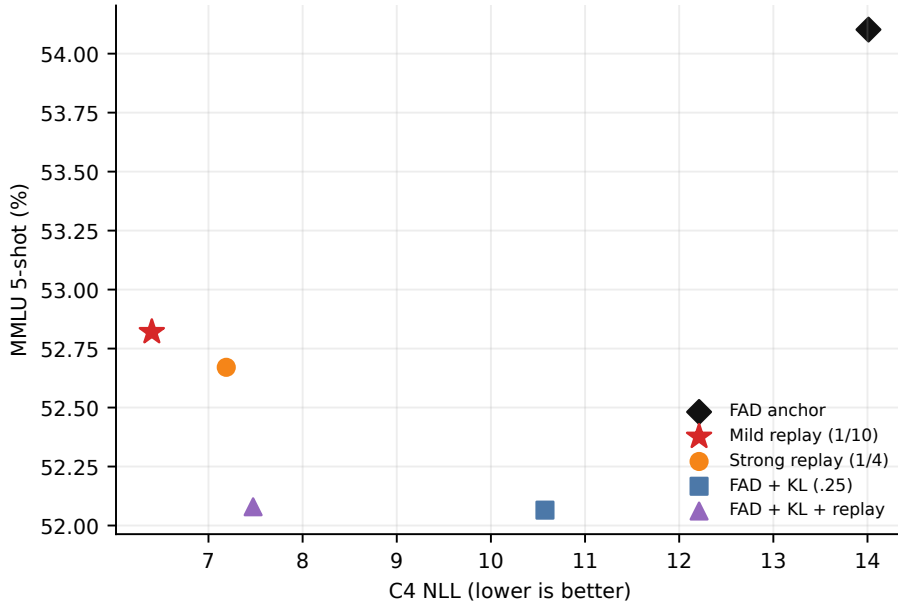


Figure E.1: **Generic-language trade-off under replay mitigation.** Low-frequency C4 replay substantially reduces generic-language NLL while retaining most of the five-shot MMLU performance.

F FUNCTIONAL SHARING CONSTRUCTION

F.1 FIXED-BUDGET ASSIGNMENT CONTROL

Table F.1: Sharing-policy comparison at fixed $K = 19$ capacity. The recovery objective, seed, and evaluation protocol are fixed; only the assignment changes.

Sharing policy	K	Commonsense macro
Restricted-late	19	85.30
Random	19	82.52
Unrestricted complete-link	19	80.36
Same-regime complete-link	19	80.32
Adjacent grouping	19	65.74

At fixed capacity, assignment choice materially affects recovery. This table is a control for the structural testbed, not a second main contribution: all controlled objective comparisons in the paper freeze the same restricted-late assignment.

F.2 REPLACEMENT COST

For an admissible pair (ℓ, ℓ') , temporarily replace the FFN at layer ℓ' with that at layer ℓ while keeping the remaining model fixed:

$$C_{\ell \rightarrow \ell'} = \Delta \text{NLL}_{\ell \rightarrow \ell'} + \lambda_{\text{KL}} D_{\text{KL}}(p_{\text{base}} \| p_{\ell \rightarrow \ell'}), \quad \lambda_{\text{KL}} = 1. \quad (\text{F.1})$$

The symmetric cost uses the worse direction,

$$C_{\ell \ell'}^{\text{bi}} = \max(C_{\ell \rightarrow \ell'}, C_{\ell' \rightarrow \ell}). \quad (\text{F.2})$$

Complete-link merging controls the largest cross-pair cost; restricted-late sharing additionally applies the admissibility mask used by the primary run.

F.3 COMPLETE-LINK CRITERION AND ITS SCOPE

For two current clusters A and B , complete-link agglomeration uses

$$d_{\text{CL}}(A, B) = \max_{\ell \in A, \ell' \in B} C_{\ell \ell'}^{\text{bi}}. \quad (\text{F.3})$$

After merging A and B , the diameter of the resulting cluster $G = A \cup B$ is

$$\text{diam}_G(G) = \max_{i, j \in G} C_{ij}^{\text{bi}}. \quad (\text{F.4})$$

Thus complete-link directly tracks a worst observed pairwise replacement cost, rather than an average pairwise cost. Cutting the hierarchy at a merge height τ produces clusters whose pairwise diameter does not exceed τ among the admissible pairs represented in the hierarchy. This is the formal reason for using complete-link when a single high-cost pairing inside a shared group is undesirable.

The guarantee is limited to the measured surrogate. Each $C_{\ell \rightarrow \ell'}$ is obtained by replacing one layer while the remainder of the model is fixed. Simultaneously sharing several layers can create interaction terms that are absent from all single-replacement measurements. Therefore Eq. (F.4) does not upper-bound the end-to-end degradation of the final multi-layer shared model without an additional additivity or stability assumption. The fixed-budget assignment control in Appendix F.1 is the empirical validation of this surrogate.

F.4 NESTED ASSIGNMENTS

All unlisted layers remain private:

$$\begin{aligned} K = 22 : & \quad \{19, \dots, 24\}, \{25, 26\}, \\ K = 19 : & \quad \{16, 17, 18\}, \{19, \dots, 26\}, \\ K = 17 : & \quad \{14, \dots, 18\}, \{19, \dots, 26\}, \\ K = 15 : & \quad \{12, \dots, 18\}, \{19, \dots, 26\}. \end{aligned}$$

These assignments satisfy the refinement condition used by Theorem 9.

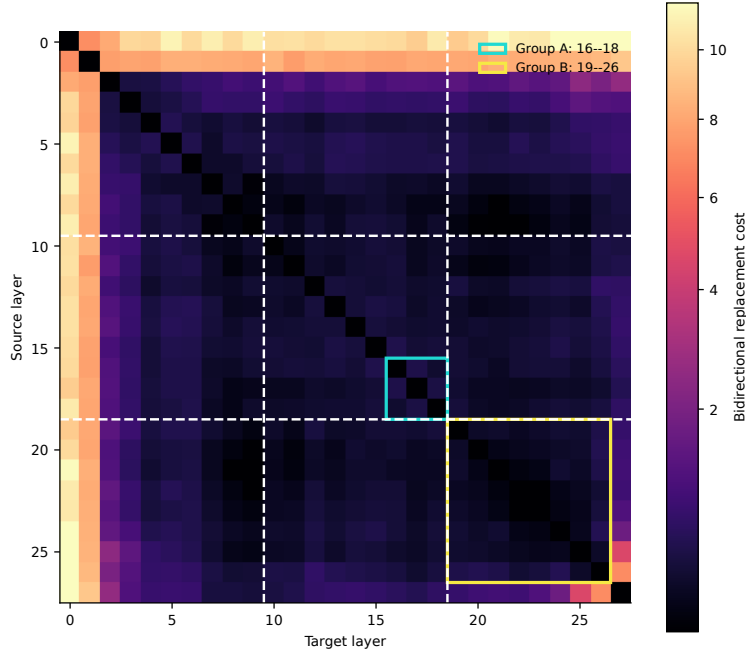


Figure F.1: **Replacement-cost geometry.** The bidirectional cost matrix $C_{\ell\ell'}^{\text{bi}}$ is shown with depth boundaries and the primary $K = 19$ restricted-late groups overlaid.

F.5 INITIALIZATION AND PRIVATE ADAPTERS

Each shared bank is initialized from its policy medoid. The private adapter is zero-initialized so that step zero isolates the mismatch introduced by sharing. Appendix C reports calibration, projection-rank, and initialization sensitivity.

F.6 CAPACITY AND INITIALIZATION OF THE PRIVATE ADAPTER

For a fixed layer and a batch of normalized FFN inputs collected as columns of $X_\ell \in \mathbb{R}^{D \times N}$, the private adapter produces

$$Y_{\text{ad}}^{(\ell)} = U^{(\ell)} V^{(\ell)} X_\ell. \quad (\text{F.5})$$

Its batch output satisfies

$$\text{rank}(Y_{\text{ad}}^{(\ell)}) \leq \min\{\rho, \text{rank}(X_\ell), N\}. \quad (\text{F.6})$$

Hence the adapter can supply at most a rank- ρ linear correction on a fixed batch. This is an expressivity statement about the correction branch; FAD supervises the sum of the shared branch and the adapter and does not require the two contributions to be identifiable.

With $U^{(\ell)} = 0$ and a nonzero initialization of $V^{(\ell)}$, the adapter output is exactly zero at step zero, while gradients can first update $U^{(\ell)}$. If both factors are initialized to zero, then the first-step gradients of both factors are zero:

$$\frac{\partial \mathcal{L}}{\partial U} = G(VX)^\top = 0, \quad \frac{\partial \mathcal{L}}{\partial V} = U^\top GX^\top = 0, \quad (\text{F.7})$$

where G is the upstream gradient. The artifact must therefore record the initialization of both $U^{(\ell)}$ and $V^{(\ell)}$, not only the fact that the adapter output begins at zero.

G COMPLETE INFORMATION-PRESERVATION DERIVATION

This appendix states the population variables, the variational objective, the measure under which the conditional density is defined, the complete maximum-entropy calculation, and the precise assumptions under which the implemented FAD loss is equivalent to maximizing a variational mutual-information lower bound. It also separates three choices that play different roles: (i) the FFN residual update is the aligned variable; (ii) teacher PCA selects the modeled coordinates; and (iii) the teacher scale matrix specifies the conditional uncertainty used by the variational surrogate. The maximum-entropy argument solves the conditional model only after these choices have been fixed.

G.1 POPULATION VARIABLES AND A WELL-DEFINED CONDITIONAL DENSITY

Let $X \sim P_X$ denote a random training example, including the token position at which the velocities are extracted. For a fixed supervised layer ℓ , define the projected teacher and student random variables

$$\mathbf{Z}_T^{(\ell)} = \mathbf{z}_T^{(\ell)}(X), \quad \mathbf{Z}_{S,\theta}^{(\ell)} = \mathbf{z}_{S,\theta}^{(\ell)}(X; \theta), \quad \theta \in \Theta_{\text{shared}}. \quad (\text{G.1})$$

The teacher, projection basis, centering vector, and scale matrix are fixed; only the student random variable depends on θ .

Because language-model inputs are discrete and both networks are deterministic, the induced distributions can be atomic. Writing a differential entropy and a Lebesgue density for such variables without qualification can therefore create a base-measure mismatch. We make the variational statement rigorous by introducing an arbitrarily small, student-independent smoothing of the teacher code,

$$\tilde{\mathbf{Z}}_{T,\tau}^{(\ell)} = \mathbf{Z}_T^{(\ell)} + \tau \boldsymbol{\varepsilon}, \quad \boldsymbol{\varepsilon} \sim \mathcal{N}(\mathbf{0}, I_r), \quad \tau > 0, \quad (\text{G.2})$$

where $\boldsymbol{\varepsilon}$ is independent of X . Conditional on any student code, the smoothed teacher variable has a density with respect to Lebesgue measure. Section G.6 shows that, for every $\tau > 0$, this smoothing changes only a student-independent constant and leaves the optimized FAD loss exactly unchanged. The construction is therefore a measure-theoretic device rather than an additional training perturbation. It is also consistent with known caveats for mutual information in deterministic neural representations (Amjad & Geiger, 2020).

Let $p_{\ell,\theta}^{(\tau)}(\tilde{\mathbf{z}}, \mathbf{s})$ denote the joint law of $(\tilde{\mathbf{Z}}_{T,\tau}^{(\ell)}, \mathbf{Z}_{S,\theta}^{(\ell)})$. We assume finite second moments and $\Sigma_T^{(\ell)} \succ 0$, which is guaranteed by the stabilizer in Eq. (4).

G.2 EXACT BILEVEL VARIATIONAL OBJECTIVE

For a realized student code $\mathbf{s} \in \mathbb{R}^r$, define the admissible conditional family

$$\begin{aligned} \mathcal{Q}_\ell(\mathbf{s}) = \Big\{ & q(\cdot | \mathbf{s}) : q(\tilde{\mathbf{z}} | \mathbf{s}) \geq 0, \quad \int q(\tilde{\mathbf{z}} | \mathbf{s}) d\tilde{\mathbf{z}} = 1, \\ & \int \tilde{\mathbf{z}} q(\tilde{\mathbf{z}} | \mathbf{s}) d\tilde{\mathbf{z}} = \mathbf{s}, \\ & \int (\tilde{\mathbf{z}} - \mathbf{s})(\tilde{\mathbf{z}} - \mathbf{s})^\top q(\tilde{\mathbf{z}} | \mathbf{s}) d\tilde{\mathbf{z}} = \Sigma_T^{(\ell)} \Big\}. \end{aligned} \quad (\text{G.3})$$

The inner solution is selected pointwise in $\mathbf{s}$ by

$$q_{\theta,\ell}^*(\cdot | \mathbf{s}) = \arg \max_{q(\cdot | \mathbf{s}) \in \mathcal{Q}_\ell(\mathbf{s})} h(q(\cdot | \mathbf{s})), \quad (\text{G.4})$$

where $h(q) = -\int q \log q$ is differential entropy. Dependence on θ appears through the realized center $\mathbf{s} = \mathbf{Z}_{S,\theta}^{(\ell)}$; there is no separately trained inference network.

The outer information-preservation score is

$$\begin{aligned} \mathcal{J}_\tau(\theta) = \frac{1}{|\mathcal{S}|} \sum_{\ell \in \mathcal{S}} & \left[I_{p_{\ell,\theta}^{(\tau)}} \left(\tilde{\mathbf{Z}}_{T,\tau}^{(\ell)}; \mathbf{Z}_{S,\theta}^{(\ell)} \right) \right. \\ & \left. - \mathbb{E}_{p_{\ell,\theta}^{(\tau)}(\mathbf{s})} D_{\text{KL}} \left(p_{\ell,\theta}^{(\tau)}(\tilde{\mathbf{z}} | \mathbf{s}) \parallel q_{\theta,\ell}^*(\tilde{\mathbf{z}} | \mathbf{s}) \right) \right]. \end{aligned} \quad (\text{G.5})$$

The nonnegative KL term gives the exact gap

$$\begin{aligned} \frac{1}{|\mathcal{S}|} \sum_{\ell \in \mathcal{S}} I\left(\tilde{\mathbf{Z}}_{T,\tau}^{(\ell)}; \mathbf{Z}_{S,\theta}^{(\ell)}\right) - \mathcal{J}_\tau(\theta) \\ = \frac{1}{|\mathcal{S}|} \sum_{\ell \in \mathcal{S}} \mathbb{E}_{p(\mathbf{s})} D_{\text{KL}}(p(\tilde{\mathbf{z}} | \mathbf{s}) \| q_{\theta,\ell}^*(\tilde{\mathbf{z}} | \mathbf{s})) \geq 0. \end{aligned} \quad (\text{G.6})$$

Consequently, FAD maximizes a specified variational lower bound. It does not claim that the true posterior belongs to the fixed-covariance Gaussian family, and it does not directly estimate the unknown gap in Eq. (G.6).

G.3 BARBER–AGAKOV REARRANGEMENT, STEP BY STEP

Fix ℓ and suppress layer and parameter subscripts. Under the smoothed population law,

$$\begin{aligned} I(\tilde{\mathbf{Z}}_T; \mathbf{Z}_S) &= \mathbb{E}_p \left[\log \frac{p(\tilde{\mathbf{z}} | \mathbf{s})}{p(\tilde{\mathbf{z}})} \right] \\ &= h_p(\tilde{\mathbf{Z}}_T) + \mathbb{E}_p[\log p(\tilde{\mathbf{z}} | \mathbf{s})]. \end{aligned} \quad (\text{G.7})$$

The conditional KL term is

$$\mathbb{E}_{p(\mathbf{s})} D_{\text{KL}}(p(\tilde{\mathbf{z}} | \mathbf{s}) \| q(\tilde{\mathbf{z}} | \mathbf{s})) = \mathbb{E}_p[\log p(\tilde{\mathbf{z}} | \mathbf{s})] - \mathbb{E}_p[\log q(\tilde{\mathbf{z}} | \mathbf{s})]. \quad (\text{G.8})$$

Subtracting Eq. (G.8) from Eq. (G.7) cancels the intractable true conditional term and yields the Barber–Agakov identity

$$I(\tilde{\mathbf{Z}}_T; \mathbf{Z}_S) - \mathbb{E}_{p(\mathbf{s})} D_{\text{KL}}(p(\tilde{\mathbf{z}} | \mathbf{s}) \| q(\tilde{\mathbf{z}} | \mathbf{s})) = h_p(\tilde{\mathbf{Z}}_T) + \mathbb{E}_p[\log q(\tilde{\mathbf{z}} | \mathbf{s})]. \quad (\text{G.9})$$

The teacher marginal entropy in Eq. (G.9) is independent of θ . Hence the student-dependent part of the lower bound is exactly the conditional log score assigned by q to paired teacher–student codes. This identity, rather than an explicit numerical estimator of mutual information, is what is optimized in the implementation (Barber & Agakov, 2003; Poole et al., 2019).

G.4 FUNCTIONAL LAGRANGIAN FOR THE CONDITIONAL MAXIMUM-ENTROPY PROBLEM

Fix a conditioning value $\mathbf{s}$, set $\mathbf{y} = \tilde{\mathbf{z}} - \mathbf{s}$, and write $\Sigma = \Sigma_T^{(\ell)}$. The inner problem becomes

$$\begin{aligned} \max_q \quad & - \int q(\mathbf{y}) \log q(\mathbf{y}) d\mathbf{y}, \\ \text{s.t.} \quad & \int q(\mathbf{y}) d\mathbf{y} = 1, \quad \int \mathbf{y} q(\mathbf{y}) d\mathbf{y} = \mathbf{0}, \\ & \int \mathbf{y} \mathbf{y}^\top q(\mathbf{y}) d\mathbf{y} = \Sigma. \end{aligned} \quad (\text{G.10})$$

Introduce a scalar multiplier α , a vector multiplier $\boldsymbol{\beta}$, and a symmetric matrix multiplier Λ . A functional Lagrangian is

$$\begin{aligned} \mathfrak{L}[q, \alpha, \boldsymbol{\beta}, \Lambda] &= - \int q \log q + \alpha \left(\int q - 1 \right) + \boldsymbol{\beta}^\top \int \mathbf{y} q \\ &\quad + \text{tr} \left[\Lambda \left(\int \mathbf{y} \mathbf{y}^\top q - \Sigma \right) \right], \end{aligned} \quad (\text{G.11})$$

where all integrals are over $\mathbb{R}^T$. For an arbitrary integrable perturbation δq , the first variation is

$$\delta \mathfrak{L} = \int \left[-1 - \log q(\mathbf{y}) + \alpha + \boldsymbol{\beta}^\top \mathbf{y} + \mathbf{y}^\top \Lambda \mathbf{y} \right] \delta q(\mathbf{y}) d\mathbf{y}. \quad (\text{G.12})$$

A stationary density must therefore satisfy

$$q(\mathbf{y}) = \exp \left(\alpha - 1 + \boldsymbol{\beta}^\top \mathbf{y} + \mathbf{y}^\top \Lambda \mathbf{y} \right). \quad (\text{G.13})$$

Normalizability requires the quadratic term to be negative definite. Define $A = -2\Lambda \succ 0$ and $\boldsymbol{\eta} = \boldsymbol{\beta}$. Completing the square gives

$$\begin{aligned} q(\mathbf{y}) &\propto \exp\left[-\frac{1}{2}\mathbf{y}^\top A\mathbf{y} + \boldsymbol{\eta}^\top \mathbf{y}\right] \\ &\propto \exp\left[-\frac{1}{2}(\mathbf{y} - A^{-1}\boldsymbol{\eta})^\top A(\mathbf{y} - A^{-1}\boldsymbol{\eta})\right]. \end{aligned} \quad (\text{G.14})$$

Thus the stationary density is Gaussian with mean $A^{-1}\boldsymbol{\eta}$ and covariance A^{-1} . The zero-mean constraint gives $\boldsymbol{\eta} = \mathbf{0}$, and the covariance constraint gives $A = \Sigma^{-1}$. Translating back from $\mathbf{y}$ to $\tilde{\mathbf{z}}$ yields

$$q_{\theta, \ell}^*(\tilde{\mathbf{z}} \mid \mathbf{s}) = \frac{\exp\left[-\frac{1}{2}(\tilde{\mathbf{z}} - \mathbf{s})^\top (\Sigma_T^{(\ell)})^{-1}(\tilde{\mathbf{z}} - \mathbf{s})\right]}{(2\pi)^{r/2} \det(\Sigma_T^{(\ell)})^{1/2}} = \mathcal{N}(\tilde{\mathbf{z}}; \mathbf{s}, \Sigma_T^{(\ell)}). \quad (\text{G.15})$$

G.5 FEASIBILITY, STRICT CONCAVITY, AND GLOBAL UNIQUENESS

Theorem 1 (Unique conditional maximum-entropy solution). *For every $\mathbf{s} \in \mathbb{R}^r$ and every $\Sigma \succ 0$, the family in Eq. (G.3) is nonempty, convex, and has the unique entropy maximizer $\mathcal{N}(\mathbf{s}, \Sigma)$, up to equality almost everywhere.*

Proof. The Gaussian in Eq. (G.15) is feasible, so the set is nonempty. Nonnegativity and normalization define a convex set, while the mean and second-central-moment constraints are affine in q for fixed $\mathbf{s}$; therefore the intersection is convex. Differential entropy is strictly concave on densities whenever the relevant integrals exist.

A direct global proof follows from KL nonnegativity. Let $q^* = \mathcal{N}(\mathbf{s}, \Sigma)$ and let q be any feasible density. Since

$$\log q^*(\tilde{\mathbf{z}}) = -\frac{r}{2} \log(2\pi) - \frac{1}{2} \log \det \Sigma - \frac{1}{2}(\tilde{\mathbf{z}} - \mathbf{s})^\top \Sigma^{-1}(\tilde{\mathbf{z}} - \mathbf{s}), \quad (\text{G.16})$$

the shared mean and covariance constraints imply $\int q \log q^* = \int q^* \log q^* = -h(q^*)$. Hence

$$D_{\text{KL}}(q \parallel q^*) = \int q \log q - \int q \log q^* = h(q^*) - h(q) \geq 0. \quad (\text{G.17})$$

Thus $h(q) \leq h(q^*)$, with equality if and only if $q = q^*$ almost everywhere. $\square$

The theorem proves a unique global optimum for the inner density problem. It does not prove that the neural student-training problem is convex; that distinction is made explicit in Section H.4.

G.6 EXACT REDUCTION TO THE IMPLEMENTED FAD LOSS

For one paired clean sample $(\mathbf{z}_T, \mathbf{z}_S)$, the negative log density of the smoothed target is

$$\begin{aligned} -\log q^*(\tilde{\mathbf{z}}_T \mid \mathbf{z}_S) &= \frac{r}{2} \log(2\pi) + \frac{1}{2} \log \det \Sigma_T^{(\ell)} \\ &\quad + \frac{1}{2}(\tilde{\mathbf{z}}_T - \mathbf{z}_S)^\top (\Sigma_T^{(\ell)})^{-1}(\tilde{\mathbf{z}}_T - \mathbf{z}_S). \end{aligned} \quad (\text{G.18})$$

Using $\tilde{\mathbf{z}}_T = \mathbf{z}_T + \tau \boldsymbol{\varepsilon}$ and independence,

$$\begin{aligned} \mathbb{E}_{\boldsymbol{\varepsilon}} [(\tilde{\mathbf{z}}_T - \mathbf{z}_S)^\top \Sigma^{-1}(\tilde{\mathbf{z}}_T - \mathbf{z}_S)] \\ = (\mathbf{z}_T - \mathbf{z}_S)^\top \Sigma^{-1}(\mathbf{z}_T - \mathbf{z}_S) + \tau^2 \text{tr}(\Sigma^{-1}). \end{aligned} \quad (\text{G.19})$$

The second term is fixed with respect to θ . Consequently, for fixed teacher geometry, projection rank, supervised layers, and $\tau > 0$,

$$\mathcal{J}_\tau(\theta) = C_{T, \tau} - \frac{1}{2} \mathcal{L}_{\text{FAD}}(\theta), \quad (\text{G.20})$$

where

$$C_{T,\tau} = \frac{1}{|\mathcal{S}|} \sum_{\ell \in \mathcal{S}} \left[h\left(\tilde{\mathbf{Z}}_{T,\tau}^{(\ell)}\right) - \frac{r}{2} \log(2\pi) - \frac{1}{2} \log \det \Sigma_T^{(\ell)} - \frac{\tau^2}{2} \text{tr}((\Sigma_T^{(\ell)})^{-1}) \right] \quad (\text{G.21})$$

and

$$\mathcal{L}_{\text{FAD}}(\theta) = \frac{1}{|\mathcal{S}|} \sum_{\ell \in \mathcal{S}} \mathbb{E}_X \left[\left\| (\Sigma_T^{(\ell)})^{-1/2} \left(\mathbf{Z}_{S,\theta}^{(\ell)} - \mathbf{Z}_T^{(\ell)} \right) \right\|_2^2 \right]. \quad (\text{G.22})$$

Equation (G.20) proves that minimizing the implemented clean FAD loss is exactly equivalent to maximizing the smoothed variational lower bound; no noise needs to be sampled during training.

Remark 1 (Constants are fixed only within one geometry). Dropping the Gaussian normalization, log determinant, teacher entropy, and smoothing trace is valid when $\mathbf{B}$, r , $c(\ell)$, $\Sigma_T^{(\ell)}$, ϵ , and $\mathcal{S}$ are fixed. Raw FAD values obtained with different ranks, covariance estimators, coordinate groups, or stabilizers do not by themselves represent differences between complete variational lower bounds, because the constants in Eq. (G.21) also change. This is why the nested-budget analysis in Appendix D explicitly fixes the teacher geometry.

G.7 EMPIRICAL ESTIMATOR AND SAMPLING STATEMENT

For N paired examples, the implemented sample objective is

$$\hat{\mathcal{L}}_{\text{FAD}} = \frac{1}{N|\mathcal{S}|} \sum_{n=1}^N \sum_{\ell \in \mathcal{S}} \left\| (\Sigma_T^{(\ell)})^{-1/2} (\mathbf{z}_{S,n}^{(\ell)} - \mathbf{z}_{T,n}^{(\ell)}) \right\|_2^2. \quad (\text{G.23})$$

When the geometry is estimated on an independent calibration split and then held fixed, Eq. (G.23) is the ordinary sample-average estimator of Eq. (G.22). If the same examples are used to estimate the geometry and to compute the loss, it is instead a plug-in objective; freezing the geometry still makes student optimization well defined, but a population concentration statement must condition on or account for the data-dependent geometry.

Proposition 2 (A basic fixed-geometry concentration bound). *Assume the per-example layer average*

$$D(X) = \frac{1}{|\mathcal{S}|} \sum_{\ell \in \mathcal{S}} \left\| (\Sigma_T^{(\ell)})^{-1/2} (\mathbf{Z}_{S,\theta}^{(\ell)} - \mathbf{Z}_T^{(\ell)}) \right\|_2^2 \quad (\text{G.24})$$

is bounded in $[0, M]$ for fixed θ and fixed geometry. Then, with probability at least $1 - \delta$ over N independent examples,

$$\left| \hat{\mathcal{L}}_{\text{FAD}} - \mathcal{L}_{\text{FAD}} \right| \leq M \sqrt{\frac{\log(2/\delta)}{2N}}. \quad (\text{G.25})$$

Proof. Apply Hoeffding’s inequality to the independent bounded variables $D(X_n)$. $\square$

The boundedness assumption can be replaced by a sub-exponential tail assumption, or enforced diagnostically through clipping for the purpose of reporting a robust confidence interval. The proposition is not used to claim a generalization bound for downstream accuracy.

G.8 OVERALL STUDENT OBJECTIVE AND THE ROLE OF THE TRADE-OFF WEIGHT

With the fixed structural class made explicit, the population training problem is

$$\theta_S^* \in \arg \min_{\theta \in \Theta_{\text{shared}}(K, g, \rho)} \{ \mathcal{L}_{\text{CE}}(\theta) + \lambda_{\text{FAD}} \mathcal{L}_{\text{FAD}}(\theta) \}. \quad (\text{G.26})$$

By Eq. (G.20), this is equivalent, up to the fixed constant $2\lambda_{\text{FAD}} C_{T,\tau}$, to

$$\underset{\theta \in \Theta_{\text{shared}}(K, g, \rho)}{\text{minimize}} \quad \mathcal{L}_{\text{CE}}(\theta) - 2\lambda_{\text{FAD}} \mathcal{J}_\tau(\theta). \quad (\text{G.27})$$

Thus λ_{FAD} controls the trade-off between task supervision and the variational teacher-transport score. It does not change the inner maximum-entropy solution. The factor 2 is conventional and may be absorbed into λ_{FAD} .

Remark 2 (What is imposed and what is derived). The information derivation does not independently derive the PCA basis or the teacher covariance from mutual information. The aligned variable, coordinate map, identity conditional mean, and teacher scale are modeling choices. Given those choices, maximum entropy uniquely derives the Gaussian conditional, and the Barber–Agakov identity then derives the Mahalanobis loss.

Remark 3 (Identity conditional mean). A broader variational family could use $q(z_T | z_S) = \mathcal{N}(Az_S + b, \Sigma)$ and learn an auxiliary decoder. FAD deliberately fixes $A = I$ and $b = 0$. Therefore the student code itself, not a post-hoc reparameterization of that code, must predict the teacher transport in the shared teacher coordinate system.

Remark 4 (Fixed covariance as a scale matrix). $\Sigma_T^{(\ell)}$ is not asserted to be the unknown true conditional covariance of $p(z_T | z_S)$. It is a fixed teacher-induced scale constraint. Keeping it fixed prevents the student from weakening the alignment penalty by inflating a learned variance. The resulting inverse-variance weighting measures errors in teacher-standardized units within the retained coordinates.

H THEORETICAL PROPERTIES OF GEOMETRY AND TRAJECTORY PRESERVATION

H.1 THE INDUCED METRIC IN AMBIENT VELOCITY SPACE

Fix one layer, write $B = B_{c(\ell)}$, and define the ambient velocity error $\delta_v = v_S^{(\ell)} - v_T^{(\ell)}$. Since the same teacher centroid is applied to both models,

$$z_S^{(\ell)} - z_T^{(\ell)} = B^\top \delta_v. \quad (\text{H.1})$$

Thus the centroid affects coordinate estimation but cancels exactly from each paired alignment residual.

Proposition 3 (Low-rank ambient metric). *The per-layer FAD distortion can be written as*

$$\begin{aligned} d_{\text{FAD}}^{(\ell)} &= \left\| (\Sigma_T^{(\ell)})^{-1/2} B^\top \delta_v \right\|_2^2 \\ &= \delta_v^\top \underbrace{B(\Sigma_T^{(\ell)})^{-1} B^\top}_{M_T^{(\ell)} \succeq 0} \delta_v. \end{aligned} \quad (\text{H.2})$$

$M_T^{(\ell)}$ is positive semidefinite and has rank at most r . If $\Sigma_T^{(\ell)} \succ 0$ and B has r orthonormal columns, its rank is exactly r .

Proof. Substitute Eq. (H.1) into the quadratic form. For any $a \in \mathbb{R}^D$,

$$a^\top M_T^{(\ell)} a = \|(\Sigma_T^{(\ell)})^{-1/2} B^\top a\|_2^2 \geq 0.$$

The rank statement follows from the full column rank of B and invertibility of $\Sigma_T^{(\ell)}$. $\square$

Let $P_B = BB^\top$. The projected error obeys

$$\frac{\|B^\top \delta_v\|_2^2}{\lambda_{\max}(\Sigma_T^{(\ell)})} \leq d_{\text{FAD}}^{(\ell)} \leq \frac{\|B^\top \delta_v\|_2^2}{\lambda_{\min}(\Sigma_T^{(\ell)})}. \quad (\text{H.3})$$

FAD therefore controls a norm on the retained subspace, but it is blind to the nullspace of $B^\top$. More precisely,

$$\begin{aligned} \|\delta_v\|_2^2 &= \|B^\top \delta_v\|_2^2 + \|(I - P_B)\delta_v\|_2^2 \\ &\leq \lambda_{\max}(\Sigma_T^{(\ell)}) d_{\text{FAD}}^{(\ell)} + \underbrace{\|(I - P_B)\delta_v\|_2^2}_{\text{off-subspace leakage}}. \end{aligned} \quad (\text{H.4})$$

Accordingly, a theorem about full D -dimensional velocity control would also require the off-subspace leakage to be measured or bounded. The empirical diagnostics in Appendix C therefore support directional fidelity without claiming that the projected loss alone controls every ambient direction.

H.2 WHY TEACHER PCA IS THE OPTIMAL RANK- r TEACHER SUBSPACE

For coordinate group c , let $\mathbf{Y}_c = \mathbf{v}_T - \boldsymbol{\mu}_c$ denote the centered teacher velocity under the mixture of calibration samples and layers assigned to that group, and let $C_c = \mathbb{E}[\mathbf{Y}_c \mathbf{Y}_c^\top]$. For any orthonormal $W \in \mathbb{R}^{D \times r}$,

$$\mathbb{E} \|\mathbf{Y}_c - WW^\top \mathbf{Y}_c\|_2^2 = \text{tr}(C_c) - \text{tr}(W^\top C_c W). \quad (\text{H.5})$$

Theorem 4 (Optimal teacher reconstruction subspace). *Let $\lambda_1(C_c) \geq \dots \geq \lambda_D(C_c) \geq 0$. The minimum of Eq. (H.5) over all orthonormal W is attained by choosing the top- r eigenvectors of C_c , and the minimum residual is*

$$\min_{W^\top W = I_r} \mathbb{E} \|\mathbf{Y}_c - WW^\top \mathbf{Y}_c\|_2^2 = \sum_{j=r+1}^D \lambda_j(C_c). \quad (\text{H.6})$$

Proof. Equation (H.5) follows by expanding the squared norm and using $WW^\top$ as an orthogonal projector. Maximizing $\text{tr}(W^\top C_c W)$ over r orthonormal columns selects the top- r eigenspace, whose captured variance is $\sum_{j=1}^r \lambda_j(C_c)$. Subtracting from $\text{tr}(C_c)$ gives Eq. (H.6). This is the covariance form of the Eckart–Young low-rank approximation result (Eckart & Young, 1936). $\square$

The theorem justifies PCA as the rank- r subspace that best reconstructs teacher velocity variation. It does not prove that the same subspace captures all teacher–student error. That stronger statement additionally requires the leakage term in Eq. (H.4) to be small, which is an empirical property to measure.

H.3 FULL COVARIANCE, DIAGONAL APPROXIMATION, AND COORDINATE INVARIANCE

Let the regularized full projected teacher covariance be

$$C_T^{(\ell)} = \text{Cov}(\mathbf{Z}_T^{(\ell)}) + \epsilon I_r, \quad D_T^{(\ell)} = \text{Diag}(\text{Cov}(\mathbf{Z}_T^{(\ell)})) + \epsilon I_r. \quad (\text{H.7})$$

The primary method uses $D_T^{(\ell)} = \Sigma_T^{(\ell)}$. If the full matrix were used in the MaxEnt constraint, the same derivation would yield the full Mahalanobis distortion. The diagonal choice is therefore an estimator and computational restriction, not a requirement of maximum entropy.

Proposition 5 (Invariance of the full metric). *For any invertible coordinate change $A \in \mathbb{R}^{r \times r}$, let $e' = Ae$ and $C' = ACA^\top$. Then*

$$(e')^\top (C')^{-1} e' = e^\top C^{-1} e. \quad (\text{H.8})$$

Proof. Use $(ACA^\top)^{-1} = A^{-\top} C^{-1} A^{-1}$. $\square$

A diagonal approximation remains invariant to coordinate permutations, sign changes, and coordinate-wise rescalings when its scale is transformed accordingly. It is not generally invariant to arbitrary rotations, which can move covariance mass into off-diagonal entries that are then discarded. This separates the role of the coordinate basis from that of the scale estimator.

The approximation error can be bounded through the regularized correlation matrix

$$R_T^{(\ell)} = (D_T^{(\ell)})^{-1/2} C_T^{(\ell)} (D_T^{(\ell)})^{-1/2}. \quad (\text{H.9})$$

If, for some $0 \leq \delta < 1$,

$$(1 - \delta)I_r \preceq R_T^{(\ell)} \preceq (1 + \delta)I_r, \quad (\text{H.10})$$

then every projected error e satisfies

$$\frac{1}{1 + \delta} e^\top (D_T^{(\ell)})^{-1} e \leq e^\top (C_T^{(\ell)})^{-1} e \leq \frac{1}{1 - \delta} e^\top (D_T^{(\ell)})^{-1} e. \quad (\text{H.11})$$

This follows by inverting the spectral bounds in Eq. (H.10). Reporting the spectrum of $R_T^{(\ell)}$, or a summary of off-diagonal correlations, would therefore quantify when the diagonal metric is a close approximation to the full one.

Under diagonal standardization, each retained coordinate is scaled by its own teacher standard deviation, but the coordinates need not become statistically independent. In particular, the standardized teacher covariance is $R_T^{(\ell)}$, not necessarily I_r .

H.4 CONVEXITY IN CODE SPACE, NONCONVEXITY IN PARAMETERS, AND CONDITIONING

For fixed target t and fixed $\Sigma \succ 0$, define

$$d(s, t) = (s - t)^\top \Sigma^{-1} (s - t). \quad (\text{H.12})$$

Its gradient and Hessian with respect to the student code are

$$\nabla_s d = 2\Sigma^{-1}(s - t), \quad \nabla_s^2 d = 2\Sigma^{-1} \succ 0. \quad (\text{H.13})$$

Thus the per-pair loss is strictly convex in s and has the unique code-space minimizer $s = t$. In ambient velocity space the Hessian is $2\mathbf{B}\Sigma^{-1}\mathbf{B}^\top \succeq 0$ and is not strictly positive definite when $r < D$, because off-subspace directions are unpenalized.

For the diagonal regularization in Eq. (4), $\lambda_{\min}(\Sigma) \geq \epsilon$, which gives

$$\|\nabla_s d\|_2 \leq \frac{2}{\epsilon} \|s - t\|_2, \quad \frac{2}{\lambda_{\max}(\Sigma)} I_r \preceq \nabla_s^2 d \preceq \frac{2}{\epsilon} I_r. \quad (\text{H.14})$$

The stabilizer therefore ensures existence of the inverse and caps the largest code-space curvature. Its value also changes the relative weight assigned to low-variance directions and must be reported.

Remark 5 (No global convexity claim for student training). The maps $s = \mathbf{z}_S(X; \theta)$, the shared FFN, and the factorized adapters are nonlinear and nonconvex in θ . Therefore Eq. (G.26) is not made globally convex by the quadratic code loss. The inner MaxEnt problem has a unique global solution; standard student optimization is only expected to find a stationary or approximately optimal point unless additional architecture-specific assumptions are proved.

H.5 WHY HIDDEN-STATE MATCHING DOES NOT IDENTIFY THE LOCAL VELOCITY

At one layer,

$$h_S^{(\ell+1)} - h_T^{(\ell+1)} = (u_S^{(\ell)} - u_T^{(\ell)}) + (\mathbf{v}_S^{(\ell)} - \mathbf{v}_T^{(\ell)}). \quad (\text{H.15})$$

A small next-state error does not imply a small velocity error, because the two terms on the right can cancel. Conversely, exact velocity alignment does not by itself remove a pre-existing post-attention state mismatch. The elementary bounds

$$\begin{aligned} \|\mathbf{v}_S^{(\ell)} - \mathbf{v}_T^{(\ell)}\|_2 &\leq \|h_S^{(\ell+1)} - h_T^{(\ell+1)}\|_2 + \|u_S^{(\ell)} - u_T^{(\ell)}\|_2, \\ \|h_S^{(\ell+1)} - h_T^{(\ell+1)}\|_2 &\leq \|u_S^{(\ell)} - u_T^{(\ell)}\|_2 + \|\mathbf{v}_S^{(\ell)} - \mathbf{v}_T^{(\ell)}\|_2 \end{aligned} \quad (\text{H.16})$$

show that either quantity controls the other only when the remaining term is also controlled. This is the formal reason that hidden-state MSE and velocity recovery are distinct objectives.

The interpolation in Eq. (2) is an exact within-block construction: its derivative equals the FFN residual update by definition. It does not assert that depth is continuous physical time or that the entire transformer is the solution of a particular ordinary differential equation.

H.6 FROM LOCAL VELOCITY ERROR TO TRAJECTORY DRIFT

Define the post-attention map

$$A_M^{(\ell)}(h) = h + \text{Attn}_M^{(\ell)}(\text{Norm}_1(h)), \quad u_M^{(\ell)} = A_M^{(\ell)}(h_M^{(\ell)}). \quad (\text{H.17})$$

Let $e_\ell = \|h_S^{(\ell)} - h_T^{(\ell)}\|_2$ and $\delta_\ell = \|\mathbf{v}_S^{(\ell)} - \mathbf{v}_T^{(\ell)}\|_2$.

Assumption 6 (Local post-attention stability). On the teacher–student states encountered by the paired data, there are finite constants $a_\ell \geq 0$ and $\alpha_\ell \geq 0$ such that

$$\begin{aligned} \|A_S^{(\ell)}(x) - A_S^{(\ell)}(y)\|_2 &\leq a_\ell \|x - y\|_2, \\ \|A_S^{(\ell)}(h_T^{(\ell)}) - A_T^{(\ell)}(h_T^{(\ell)})\|_2 &\leq \alpha_\ell. \end{aligned} \quad (\text{H.18})$$

The second term is zero when teacher and student use the same post-attention operator; freezing attention in both models is not by itself sufficient for $\alpha_\ell = 0$ unless the frozen weights are identical.

Theorem 7 (Discrete trajectory perturbation bound). *Under Assumption 6, for any $m \geq \ell$,*

$$e_{m+1} \leq \left(\prod_{j=\ell}^m a_j \right) e_\ell + \sum_{k=\ell}^m \left(\prod_{j=k+1}^m a_j \right) (\alpha_k + \delta_k), \quad (\text{H.19})$$

where an empty product equals one. If $a_j \leq a$ uniformly, then

$$e_{m+1} \leq a^{m-\ell+1} e_\ell + \sum_{k=\ell}^m a^{m-k} (\alpha_k + \delta_k). \quad (\text{H.20})$$

Proof. From the residual transition,

$$\begin{aligned} e_{\ell+1} &\leq \|A_S^{(\ell)}(h_S^{(\ell)}) - A_T^{(\ell)}(h_T^{(\ell)})\|_2 + \delta_\ell \\ &\leq a_\ell e_\ell + \alpha_\ell + \delta_\ell. \end{aligned} \quad (\text{H.21})$$

Repeated substitution gives Eq. (H.19); the uniform bound follows by replacing each product with the corresponding power of a . $\square$

To connect the theorem to FAD, define the off-subspace leakage

$$\eta_\ell = \|(I - P_{\mathcal{B}_{c(\ell)}})(\mathbf{v}_S^{(\ell)} - \mathbf{v}_T^{(\ell)})\|_2. \quad (\text{H.22})$$

Equation (H.4) gives

$$\delta_\ell \leq \sqrt{\lambda_{\max}(\Sigma_T^{(\ell)}) d_{\text{FAD}}^{(\ell)} + \eta_\ell^2}. \quad (\text{H.23})$$

Taking expectations and using Jensen’s inequality yields

$$\mathbb{E}[\delta_\ell] \leq \sqrt{\lambda_{\max}(\Sigma_T^{(\ell)}) \mathbb{E}[d_{\text{FAD}}^{(\ell)}] + \mathbb{E}[\eta_\ell^2]}. \quad (\text{H.24})$$

Substitution into Eq. (H.19) establishes a conditional route from low projected FAD distortion to low expected trajectory drift. The route requires controlled off-subspace leakage, attention mismatch, and layer amplification factors. Moreover, if FAD is applied only on $\ell \in \mathcal{S} \subsetneq [L]$, the unsupervised δ_ℓ terms remain in the bound; they cannot be deleted without an additional assumption or measurement.

H.7 LOGIT, PROBABILITY, AND PREDICTION-MARGIN CONSEQUENCES

Let R_M denote the final normalization and language-model readout, and let $a_M = R_M(h_M^{(L+1)})$ be the logits at the evaluated token. Suppose, on the paired states,

$$\begin{aligned} \|R_S(x) - R_S(y)\|_2 &\leq L_R \|x - y\|_2, \\ \|R_S(h_T^{(L+1)}) - R_T(h_T^{(L+1)})\|_2 &\leq \beta_R. \end{aligned} \quad (\text{H.25})$$

Then

$$\|a_S - a_T\|_2 \leq L_R e_{L+1} + \beta_R. \quad (\text{H.26})$$

For a shared final normalization and LM head, $\beta_R = 0$. Since the softmax Jacobian is $\text{Diag}(p) - pp^\top$ and has spectral norm at most $1/2$,

$$\|\text{softmax}(a_S) - \text{softmax}(a_T)\|_2 \leq \frac{1}{2} \|a_S - a_T\|_2. \quad (\text{H.27})$$

Proposition 8 (Teacher-margin preservation). *Let $j_T = \arg \max_j a_{T,j}$ and define the teacher logit margin*

$$\gamma_T = a_{T,j_T} - \max_{j \neq j_T} a_{T,j}. \quad (\text{H.28})$$

If $\|a_S - a_T\|_\infty < \gamma_T/2$, then the student and teacher have the same argmax prediction.

Proof. The winning teacher logit can decrease by at most $\|a_S - a_T\|_\infty$, while any competing logit can increase by at most the same amount. A perturbation smaller than half the margin cannot reverse their order. $\square$

This proposition explains why a trajectory bound can support prediction preservation on high-margin examples. Accuracy is nevertheless discontinuous, and no unconditional accuracy guarantee follows without a margin distribution and verified constants.

H.8 NESTED SHARING CLASSES AND THE EXACT MONOTONICITY CLAIM

Let Π_K denote the layer partition associated with K stored FFN banks, and let Θ_K be the corresponding functional model class, including the same per-layer adapter class. A partition Π_{K_2} is a refinement of Π_{K_1} when every group in Π_{K_2} is contained in a group in Π_{K_1} .

Theorem 9 (Nested functional classes). *Assume $K_2 > K_1$, Π_{K_2} refines Π_{K_1} , shared-bank weights are trainable without inequality constraints that prevent two banks from being equal, and the adapter class is unchanged. Then*

$$\Theta_{K_1} \subseteq \Theta_{K_2}. \quad (\text{H.29})$$

Consequently,

$$\begin{aligned} \inf_{\theta \in \Theta_{K_2}} \mathcal{L}_{\text{FAD}}(\theta) &\leq \inf_{\theta \in \Theta_{K_1}} \mathcal{L}_{\text{FAD}}(\theta), \\ \sup_{\theta \in \Theta_{K_2}} \mathcal{J}_\tau(\theta) &\geq \sup_{\theta \in \Theta_{K_1}} \mathcal{J}_\tau(\theta), \\ \inf_{\theta \in \Theta_{K_2}} (\mathcal{L}_{\text{CE}} + \lambda_{\text{FAD}} \mathcal{L}_{\text{FAD}}) &\leq \inf_{\theta \in \Theta_{K_1}} (\mathcal{L}_{\text{CE}} + \lambda_{\text{FAD}} \mathcal{L}_{\text{FAD}}). \end{aligned} \quad (\text{H.30})$$

Proof. Any model in Θ_{K_1} is represented in Θ_{K_2} by copying the same coarse-group bank weights into every refined subgroup and retaining the same private adapters. Taking an infimum over a superset cannot increase the optimum loss, and taking a supremum over a superset cannot decrease the optimum score. The second line also follows from Eq. (G.20) under fixed geometry. $\square$

The restricted-late assignments in Appendix F satisfy the required refinement pattern if implemented exactly as listed. Medoid initialization affects the optimization path but not the model-class inclusion, provided the banks remain trainable.

Remark 6 (What monotonicity does not guarantee). The theorem concerns global optima. It does not guarantee that independently trained nonconvex checkpoints attain those optima. In addition, the FAD component at a minimizer of the combined CE–FAD objective need not be monotone even though the optimum value of the combined objective is monotone. Therefore a monotone sequence of observed FAD losses is empirical support for the nested-class prediction, not a proof of optimizer convergence.

H.9 STRUCTURAL RATE–DISTORTION INTERPRETATION

Let $B(\theta)$ denote a structural storage cost, such as the number of stored FFN parameters including private adapters. Define the best achievable transport distortion under budget b by

$$D^*(b) = \inf_{\theta: B(\theta) \leq b} \mathcal{L}_{\text{FAD}}(\theta). \quad (\text{H.31})$$

If $b_1 \leq b_2$, then the feasible set at b_1 is contained in that at b_2 and $D^*(b_1) \geq D^*(b_2)$. Equivalently, the optimal distortion is nondecreasing as the compression fraction increases.

For a fixed structural budget b , the constrained recovery problem

$$\begin{aligned} \min_{\theta \in \Theta_b} \mathcal{L}_{\text{CE}}(\theta) \\ \text{s.t. } \mathcal{L}_{\text{FAD}}(\theta) \leq D_0 \end{aligned} \quad (\text{H.32})$$

has the Lagrangian

$$\mathcal{L}_{\text{CE}}(\theta) + \lambda(\mathcal{L}_{\text{FAD}}(\theta) - D_0), \quad \lambda \geq 0, \quad (\text{H.33})$$

whose student-dependent part is the implemented CE–FAD objective. This gives a structural rate–distortion analogue: storage is the rate-like resource and the teacher-standardized transport loss is the distortion. It should not be called a literal Shannon rate–distortion function unless a bit-level code length or a mutual-information rate is defined. Likewise, the classical information bottleneck constrains representation information rather than parameter count (Tishby et al., 2000). Because the neural problem and discrete sharing choices are nonconvex, strong duality is not guaranteed and varying λ_{FAD} need not recover every point on a nonconvex Pareto frontier.

H.10 SCOPE OF THE THEORETICAL CLAIMS

The results above support the following precise claims.

- For fixed coordinates and teacher scale, FAD exactly maximizes the student-dependent part of a smoothed Barber–Agakov lower bound with a unique maximum-entropy Gaussian conditional.
- The per-pair objective is strictly convex in the projected student code, but the end-to-end shared-weight student problem remains nonconvex.
- FAD controls a teacher-weighted norm only inside the retained teacher subspace. Full velocity control additionally requires small off-subspace leakage.
- Local velocity control yields a trajectory-drift bound only together with bounded post-attention amplification and operator mismatch; unsupervised layers remain explicit in the bound.
- Nested sharing classes order global optima under a refinement condition, not arbitrary trained checkpoints or non-nested assignments.

These qualifications delimit the information-preservation interpretation without weakening the implemented objective or the empirical questions tested in the main paper.